https://pubsonline.informs.org/journal/mksc

MARKETING SCIENCE
Vol. 42, No. 6, November–December 2023, pp. 1057–1079
ISSN 0732-2399 (print), ISSN 1526-548X (online)

Targeting Advertising Spending and Price on the Hotelling Line

Sridhar Moorthy,[a] Shervin Shahrokhi Tehrani[b,*]

[a] Rotman School of Management, University of Toronto, Toronto, Ontario M5S 3E6, Canada; [b] Naveen Jindal School of Management, University of Texas at Dallas, Richardson, Texas 75080

*Corresponding author

Contact: sridhar.moorthy@rotman.utoronto.ca, https://orcid.org/0000-0001-5251-9961 (SM); Shervin.Tehrani@UTDallas.edu, https://orcid.org/0000-0001-5244-9705 (SST)

Received: October 18, 2020
Revised: October 17, 2021; May 23, 2022; September 26, 2022; November 2, 2022
Accepted: November 4, 2022
Published Online in Articles in Advance: January 6, 2023

https://doi.org/10.1287/mksc.2022.1422

Abstract. Are competing firms better off with more targeting? We examine this question in the context of targeting advertising spending and price in the Hotelling model. By "more targeting," we mean expanding the scope of targeting from one of these variables to both. When the incremental cost of such expansion is small, the question boils down to whether the firms are better off targeting both advertising spending and price ("full targeting") or just one of them ("partial targeting"). We show that it is individually rational for the firms to expand the scope of targeting in such circumstances. However, their profits may or may not be higher in the resulting full targeting equilibrium. The paper identifies the conditions under which each type of targeting is collectively optimal. Targeting advertising spending only is optimal when the products are highly differentiated; however, when product differentiation is small, full targeting is optimal when the cost of advertising is small, and targeting price only is optimal when the cost of advertising is large. These results provide managerial guidance on how firms should respond to any restrictions on the scope of targeting imposed by regulators or online retail platforms. On the theoretical front, our results show that expanding the scope of targeting has benefits and costs for the firms. The benefits come from being able to align local advertising levels to local prices—what Dorfman and Steiner's theorem says firms should do. The costs come in the form of more intensified price competition, which takes different forms depending on which type of targeting is added. When price targeting is added to advertising spending targeting, it has the effect of replacing product differentiation by informational differentiation; when advertising spending targeting is added to price targeting, it has the effect of delegating control of the local price equilibrium to the weaker firm. Comparing our results with previous results suggests that targeting on *preferences* may be fundamentally different from targeting on loyalty and switching *behaviors*.

History: Yuxin Chen served as the senior editor for this article.
Funding: S. Moorthy was supported by the Social Sciences and Humanities Research Council of Canada [Grant 435-2017-0121].
Supplemental Material: The online appendix is available at https://doi.org/10.1287/mksc.2022.1422.

Keywords: customized marketing • location-based targeting • targeted competition

1. Introduction

It is no longer news to report that firms are able to target their advertising spending and prices. Twenty years ago, Chen and Iyer (2002, p. 198) were already writing that "advances in information technology … gives firms the ability to reach individual customers and to customize price and promotional efforts." Today, hardly anyone would consider that an overstatement. Indeed, one does not have to look far, either in the popular media (Vascellaro 2011, O'Connor 2014, Bruell 2019) or in academic journals (Sahni et al. 2016, Dubé et al. 2017, Nair et al. 2017), to discover new and creative ways by which firms practice targeted marketing.

What is news, perhaps, is that basic questions about targeting strategy remain unanswered. This paper deals with one of those questions. Is more targeting better than less? Or more precisely, are competing firms better off with more targeting in a Hotelling model? By "more targeting," we mean expanding the *scope* of targeting, from targeting advertising spending only or price only to targeting both advertising spending and price. In other words, if firms are already targeting advertising spending or price, should they also deploy the other variable in targeted fashion? From a cost perspective, the answer may seem obvious—after all, there are economies of scope in targeting. For example, if a firm already has granular data to enable one kind of targeting, it might as well use it to do the other kind of targeting. Similarly, the technologies used to deliver targeting on one variable may usually be marshaled to deliver targeting on the other variable.[1] For example, in Sahni et al. (2016), Dubé et al. (2017), and Nair et al. (2017), the emails/SMS messages/direct ("snail") mails bearing different prices for different consumers also served as advertising messages.[2] What is not clear is whether there are benefits from expanding the scope of targeting.

Going by the discourse in the business press (Dwoskin 2014, Bruell 2017), one would not think so. These journalistic accounts, lacking a formal model, seem to be channeling a monopoly framework, and in this framework, indeed, expanding the scope of targeting has benefits. Price targeting makes it possible for the firm to price discriminate, and advertising spending targeting allows the firm to tailor its advertising spending to prices and avoid wasteful spending; doing both, therefore, can only be better than doing one. However, what is good for a monopolist may not be good for a firm facing competition. This is already apparent in the seminal analysis of Thisse and Vives (1988) of price targeting in the Hotelling model; although competing firms have an individual incentive to pursue such targeting, when both do it, they end up worse off than before. However, their result, although suggestive, is not conclusive to our question. It speaks to what happens when firms go from no targeting to price targeting but does not say what will happen when they expand the scope of targeting from targeting advertising spending or price to targeting both advertising spending and price. Indeed, there is no role for advertising in the Thisse and Vives (1988) model; their consumers come endowed with full information about products and prices from the outset.[3] By contrast, in the model examined here, consumers start out uninformed, and they need to be exposed to a firm's advertising in order to become informed. In this context, as we will show, the prisoner's dilemma effect of price targeting is ameliorated. However, it is not eliminated. In fact, any of the three targeting strategies—targeting advertising spending only, targeting price only, and targeting advertising spending and price—can be collectively optimal. What is important then is to know the mapping from the economic environment to optimal targeting strategy. We provide such a mapping in this paper.

Our methodology is as follows. We take a standard two-firm Hotelling model and graft a standard informative advertising technology onto it. Then, we examine three targeting regimes: (i) both firms target advertising spending only, (ii) both firms target price only, and (iii) both firms target both advertising spending and price. We refer to the first two as "partial targeting" and the third as "full targeting." For each targeting regime, we first identify the equilibrium in advertising spending and price; these equilibria then define the profits each firm should expect in each targeting regime. Then, we open up the game to targeting strategy choice. This analysis shows that it is indeed individually rational for competing firms to expand the scope of targeting when the out-of-pocket costs of doing so are small. In other words, if the incremental cost of expanding the scope of targeting is small, neither partial targeting equilibrium can be sustained as an equilibrium in the absence of a priori restrictions on the scope of targeting. This does not, however, make the partial targeting regimes irrelevant. Restrictions on the scope of targeting may well be the order of the day—imposed on the firms by third parties, such as regulators (e.g., the European Data Protection Board) or online platforms (e.g., Amazon Marketplace). Managers, therefore, need to know what their stance should be with respect to such restrictions. Should they embrace them, or should they lobby against them? This requires knowing whether they are better or worse off in the partial versus full targeting equilibria.

The answer this paper gives is that any of the three targeting regimes may yield the highest equilibrium profits (abstracting from scope-expansion costs), depending on the economic environment. We focus specifically on two parameters of the model: t, the "unit transportation cost," and k, a parameter of the advertising cost function. The former does double duty in the Hotelling model; it speaks to consumers' disutility from not getting their "ideal" product, and it speaks to product differentiation—the price insensitivity of consumers in switching between the two products. The latter speaks to the marginal cost of making more consumers aware. We show that targeting advertising spending only is the optimal targeting strategy from an industry profit perspective when t is large. On the other hand, when t is small, optimal targeting strategy depends on k. When k is small—think low-CPM advertising vehicles such as online advertising—targeting both advertising spending and price is optimal, but when k is large—think high-CPM advertising vehicles such as TV—targeting price only is optimal.[4]

Our conditional assessments of the optimality of different targeting strategies stand in sharp contrast to the unconditional recommendations coming out of the Iyer et al. (2005) model. There, when the incremental cost of expanding the scope of targeting is zero, going from advertising spending only to advertising spending and price makes no difference to firms' profits, and expanding the scope of targeting from price only to advertising spending and price increases profits. In other words, both partial advertising spending targeting and full targeting are equally optimal from the industry's perspective, whereas partial price targeting is strictly worse. This raises a question. Why are the results different between the two models? We address this in Section 8 of the paper. As we discuss there, there are two key differences between the Iyer et al. (2005) model and ours. First, consumers in their model come *endowed* with *behaviors*; they are either loyal to a brand, or they are switchers. Firms are assumed to be able to identify and target those behavioral differences. In the Hotelling model, consumers come endowed with *preferences*, not behaviors. People are free to choose any product they want based on their awareness, preferences, and prices. Firms are assumed to be able to identify and target preference differences.[5] Second, although Iyer et al. (2005) assume that advertising exposure is deterministic, we assume that advertising exposure is probabilistic. Thus, in the Iyer et al. (2005) model, firms face binary advertising spending decisions with respect to each target market: whether to make all consumers in that market aware or nobody aware. This results in equilibria involving mixed strategies in both advertising spending and price. By contrast, in our model, advertising spending is a convex, increasing function of the *probability* of reaching a random consumer in the target market. Our equilibria involve pure strategies in advertising spending and mixed strategies in price.

This paper provides new insights on how targeted advertising spending works, how price targeting works, and the synergies and dissynergies between the two types of targeting. Perhaps the most basic is that competition works primarily through price; informative advertising's main role is that of enabler—making it possible for price to do what it wants to do (i.e., capture market share). The second insight is that competition works differently depending on whether price is targeted or not. When it is not targeted, firms are competing for the *marginal* segment; when it is targeted, they are competing for *every* segment. In the former, *product* differentiation mediates the intensity of competition; in the latter, *informational* differentiation mediates the intensity of competition. Informational differentiation is the differentiation created within homogeneous segments purely by advertising's probabilistic reach (Butters 1977; Grossman and Shapiro 1984; Tirole 1988; Pazgal et al. 2022, p. 359). Some consumers behave like "loyalists" simply because they have been exposed to one firm's advertising only; others behave like "switchers" because they have been exposed to both firms' advertising.[6] Finally, these two types of differentiation correspond to the dual role t plays in Hotelling's model. On the one hand, it reduces price sensitivity; on the other hand, it reduces willingness to pay. When price is not targeted, t is playing the former role; when it is targeted, t is playing the latter role.

In this context, what targeted advertising spending adds to targeted pricing is flexibility: the ability to fine-tune the sizes of the "loyalist" and "switcher" subsegments at each location in line with consumers' willingness to pay and the firms' competitive positions. This makes each firm behave as if it is implicitly targeting its "loyalists" in its stronger locations and "switchers" in its weaker locations.[7] On the other hand, when price targeting is accompanied by untargeted advertising spending, each firm behaves as if it is implicitly targeting its "loyalists" where it is weaker and "switchers" where it is stronger. Because the weaker firm has less to lose from lower prices, targeted price competition is more intense when it is accompanied by targeted advertising spending than when it is not. Similarly, given advertising spending targeting, going from untargeted price competition to targeted price competition increases the intensity of price competition but for a different reason. Now, it is because product differentiation is replaced by informational differentiation. In short, regardless of which type of targeting is added to existing targeting, price competition is intensified—this is the dissynergy to which we referred.

However, the paper also shows that the two types of targeting have synergies—profit-increasing features that each possesses, which are activated only in the other's presence. For targeted pricing, that feature is the possibility of price discriminating between different segments according to willingness to pay and competitive advantage. However, in order to realize the full benefits of such discrimination, local advertising spending should vary in step with local prices (this being the essence of the theorem of Dorfman and Steiner 1954). This is impossible to do if advertising spending is constant throughout the Hotelling line. Similarly, the advantage of targeting advertising spending is that it allows firms to vary their local awareness levels to suit the size of these markets and local prices. However, this benefit is partially neutered if prices cannot vary along the Hotelling line.

Thus, both advertising spending targeting and price targeting have benefits and costs in the form of synergies and dissynergies. The economic environment determines how those benefits and costs play out. When t is large, product differentiation is large, and the scales are tipped in favor of targeting advertising spending only; by not targeting price, the firms take advantage of the competition-dampening effects of product differentiation. On the other hand, when t is small, price competition is already intense under untargeted pricing; now, the bigger opportunity is the ability to price discriminate by location. Hence, price targeting is optimal either in conjunction with advertising spending targeting (when k is small, allowing awareness to increase with prices) or by itself (when k is large).

Our paper is most closely related to two papers in the literature: Chen and Iyer (2002) and Brahim et al. (2011). The first examines competition between two horizontally differentiated direct-marketing firms in a two-stage model. In the first stage, firms choose "addressability"—the proportion of consumers at each point on the Hotelling line who are in the firm's database—and in the second stage, they choose targeted prices. By interpreting "addressability" as the "fraction made aware by advertising," their results are interpretable as the partial price targeting equilibrium when untargeted advertising spending is chosen before prices ("sequential game form"). The second examines the partial targeting equilibrium when advertising spending is targeted but prices are not; it assumes that advertising spending and prices are chosen simultaneously ("simultaneous game form"). Neither paper examines full targeting. As a result, neither has anything to say about the optimal scope of targeting. Our work directly complements Brahim et al. (2011) because we, too, employ the simultaneous game form.[8] A hallmark of the simultaneous game form is that equilibria in pure advertising strategies exist in all targeting regimes. Moreover, these equilibria are unique. By contrast, as we discuss in the online appendix, the sequential game form is plagued by multiple equilibria in pure advertising strategies, some of which, being symmetrically asymmetric, provide no way to pick one as the right "solution" (other than by resorting to mixed-strategy equilibria). For these reasons, the simultaneous game form is the superior modeling strategy and the one we adopt throughout the paper.

The rest of the paper is organized as follows. In Section 2, we present our model and discuss the monopolist's choice of targeting strategy. As noted, the monopolist's case highlights the more obvious intuitions governing the targeting of advertising spending and price—what we teach in our MBA classes and what journalists write about when they discuss the merits of targeting. The less obvious intuitions surface only under competitive targeting. In Section 3, we examine partial advertising targeting—the simplest targeting case to consider. In Section 4, we examine the local targeted price equilibrium for given advertising levels; this equilibrium underpins the partial price targeting equilibrium in Section 5 and the full targeting equilibrium in Section 6. In Section 7, we ask whether it is optimal for firms to expand the scope of targeting, individually and collectively. Section 8 discusses the main forces driving our results, their robustness, and why they are different from Iyer et al. (2005). Finally, Section 9 concludes the paper. All proofs for the propositions in the text are in the appendix; the online appendix discusses targeting in the sequential game form.

2. Model

We start with a standard Hotelling (1929) model of two firms as in Figure 1. The line represents "product space." Firm 1's product is located at zero, and Firm 2's product is located at one. Each firm has the same constant marginal cost of producing its product, which we normalize to zero. Consumer "ideal points," indexed by x, are distributed uniformly along the line. Thus, consumers located at x ("segment x") have a reservation price $r - tx$ for a unit of Firm 1's product and reservation price $r - t(1-x)$ for a unit of Firm 2's product.[9] We denote $r - tx$ by $v_1(x)$, $r - t(1-x)$ by $v_2(x)$, and $|v_1(x) - v_2(x)| = |t(1-2x)|$ by $\Delta(x)$. Consumers in segment x buy a unit of Firm 1's (Firm 2's) product if and only if $v_1(x) - p_1 \geq \max\{v_2(x) - p_2, 0\}$ ($v_2(x) - p_2 \geq \max\{v_1(x) - p_1, 0\}$), where p_1 and p_2 are the prices of the two firms.

Consumers in segment $x = 1/2$ are indifferent between the two firms, and we will denote their common valuation by v. All other consumers have asymmetric preferences; at $x < 1/2$ ($x > 1/2$), consumers prefer Firm 1 (Firm 2) by an amount $\Delta(x)$. We introduce the terms "stronger firm at x" and "weaker firm at x"—identified by the subscripts "s" and "w," respectively—to refer to the firm that has the higher or lower valuation, respectively, at x. (With a slight abuse of notation, we will extend the strong-weak terminology to $x = 1/2$ as well.) Thus,

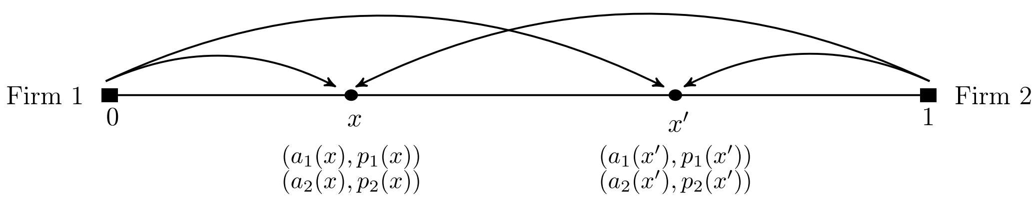
$$v_s(x) = \begin{cases} v_1(x) & \text{for } x < 1/2 \\ v & \text{for } x = 1/2 \\ v_2(x) & \text{for } x > 1/2 \end{cases}$$

$$v_w(x) = \begin{cases} v_2(x) & \text{for } x < 1/2 \\ v & \text{for } x = 1/2 \\ v_1(x) & \text{for } x > 1/2. \end{cases}$$

Finally, we will assume that $r/t > 2$ (i.e., $v_s(0) = v_s(1) < 2v_w(0) = 2v_w(1)$). With this assumption, as Chen and Iyer (2002) have noted, even "a monopoly firm [located at zero or one] will have the incentive to serve all the customers it can reach."

Consumers are unaware of the firms and uninformed about their products and prices unless reached by their advertising (Butters 1977; Grossman and Shapiro 1984; Tirole 1988, section 7.3; Chen and Iyer 2002; Iyer et al. 2005; Brahim et al. 2011; Esteves and Resende 2016). Although the most straightforward interpretation of this assumption is the literal one (namely, that these consumers really start from ground zero[10]—they have never heard of the firms and know nothing about their products and prices until exposed to their advertising), that would be unnecessarily limiting. A more useful interpretation would allow for the possibility that the consumers have heard of the firms and know something about them but that their memories are now "out of

Figure 1. Illustrating the Model



mind." Exposure to a firm's advertising *rekindles* those memories and updates them if necessary (Iyer et al. 2005).[11] This extends the scope of informative advertising to virtually any kind of advertising—even to so-called "uninformative advertising."

Whatever the interpretation, advertising exposure is probabilistic, an empirical fact first incorporated into theoretical models by Butters (1977) and since widely adopted in the informative advertising literature—including by all the papers cited except Iyer et al. (2005).[12] Advertising exposure is probabilistic because advertising sent is not the same as advertising received. Although the advertiser arguably controls the former, the latter unarguably depends on the consumer. This is obvious for ads in old-school media such as TV, radio, and print; an advertiser's TV ad, for example, no matter how well targeted may not reach a targeted consumer if the consumer chooses not to watch the program when the ad airs. However, it is also true for online advertising—the most targetable of advertising media. In this environment, exposure is a function not only of consumer unobservables but also, of platform-related factors—for example, Facebook may throttle an ad because it wants to limit the number of ads in a consumer's news feed—and other random factors related to the ad auction that ultimately determine whether and how an ad budget gets translated into ad impressions (Gordon et al. 2019). Lacking visibility into who is exposed and who is not means that firms cannot target on the basis of ad exposure.

In this framework, a firm's advertising spending level determines its probability of reaching a random consumer in the target market or in expectation, the equivalent fraction.[13] Specifically, a firm that expects to make a fraction a of a particular market segment (location) aware will need to send $(k/2)a^2$ dollars of advertising to that segment ($k > 0$).[14] This is the same advertising cost function as in Tirole (1988, p. 292), Chen and Iyer (2002), Brahim et al. (2011), and Esteves and Resende (2016). Given the one-to-one mapping between awareness and advertising spending, we will use these terms interchangeably unless there is a need to distinguish them.

A fundamental implication of advertising's probabilistic reach is that at every x, four subsegments are created depending on which firm's ads, if any, a consumer is exposed to. The sizes of these subsegments are a function of the firms' awareness levels at x, $a_1(x), a_2(x)$.

1. Consumers exposed to Firm 1's ads but not Firm 2's. The expected size of this subsegment is $a_1(x)(1 - a_2(x))$.
2. Consumers exposed to Firm 2's ads but not Firm 1's. The expected size of this subsegment is $a_2(x)(1 - a_1(x))$.
3. Consumers exposed to both firms' ads. The expected size of this subsegment is $a_1(x)a_2(x)$.
4. Consumers not reached by either firm's ads. The expected size of this subsegment is $(1 - a_1(x))(1 - a_2(x))$. These consumers, being unaware of either firm, are out of the market.

Consumers in subsegment 1 *behave as* "Firm 1 loyalists"; they will only consider Firm 1 because they are only aware of Firm 1. Similarly, consumers in subsegment 2 *behave as* "Firm 2 loyalists." Subsegment 3 consumers, however, *behave as* "switchers"; they will consider both firms and buy Firm j if and only if $v_j(x) - p_j \geq \max\{v_{3-j}(x) - p_{3-j}, 0\}$ for $j = 1, 2$.

These "loyalists" and "switchers" are different from the loyalists and switchers in the Iyer et al. (2005) model. Although in their model, they are "born that way," ours arise endogenously from differential exposure to advertising.[15] (Our model does not have any exogenous loyalists or switchers.) Most important, although their loyalists and switchers are assumed to be identifiable by the firms and hence, targetable, our "loyalists" and "switchers," being invisible to the firms, cannot be targeted.[16]

Advertising spending (price) targeting in this context means that firms have the capability, and the freedom, to choose different levels of awareness (price) for different market segments. (Whether they exercise that option is a different question.) Targeted pricing is referred to as "discriminatory pricing"[17] in the economics literature (Corts 1998), and "uniform advertising" and "random advertising" are other terms used for untargeted advertising spending (Iyer et al. 2005, Brahim et al. 2011). Firms may target only one of these variables while deploying the other in an untargeted fashion ("partial targeting"), or they may target both variables ("full targeting"). We examine three targeting regimes: (i) partial advertising targeting (only advertising spending is targeted by both firms), (ii) partial price targeting (only price is targeted by both firms), and (iii) full targeting (both advertising spending and price are targeted by both firms). Under full targeting, firms' advertising spending and price strategies are functions, $a_j : [0,1] \longmapsto [0,1]$ and $p_j : [0,1] \longmapsto [0,\infty)$, $j = 1, 2$, respectively, such that $a_j(x)$ is the jth firm's probability of making a random consumer in segment x aware of its product and $p_j(x)$ is firm j's price to segment x (see Figure 1); under partial advertising targeting, firms' advertising spending strategies are functions, but their price strategies are numbers $p_j \in [0,\infty)$, $j = 1, 2$. Finally, under partial price targeting, although firms' price strategies are functions, their advertising spending strategies are numbers $a_j \in [0,1]$, $j = 1, 2$.[18] Within each targeting regime, firms choose advertising spending and price simultaneously (i.e., each firm takes the other firm's advertising spending and price as given when choosing its own advertising spending and price). As discussed in Section 1, to account for economies of scope in targeting and to ground our comparison of different targeting regimes on the basis of the firms' profits within those regimes, we will assume that the incremental cost of expanding the scope of targeting from partial to full targeting is negligible.

How would a monopolist behave in this model? Without loss of generality, assume that the monopolist's product is located at zero. (If it was located at one, we would simply replace x with $1-x$ in the expressions.) Under partial advertising targeting, the monopolist chooses $(p, a(\cdot))$ to maximize

$$\int_0^{\min\left\{\frac{r-p}{t},1\right\}} \left(pa(x)-(k/2)a(x)^2\right)dx.$$

It follows immediately that for any p, the optimal advertising spending strategy $a(\cdot)$ must maximize local profit $pa(x)-k(a(x))^2/2$ for all $x \in \left[0, \min\left\{\frac{r-p}{t},1\right\}\right]$: that is, satisfy $a(x) \equiv \min\{p/k, 1\}$.[19] Absent variation in prices, the monopolist would not vary awareness either! Given uniform a, optimal price p must maximize

$$\left(pa-(k/2)a^2\right)\left(\min\left\{\frac{r-p}{t},1\right\}\right).$$

This yields $p=\max\{r-t,(1/2)(r+(k/2)a)$. Combining the two, the monopolist's optimal partial advertising targeting strategy is for $r<\min\{3k/2,3t\}$, $p_m^{UP}=2r/3$, $a_m^{UP}(x)\equiv 2r/3k$; for $\min\{3k/2,3t\} \le r < \max\{2t+k/2, t+k\}$ and $k \le 2t$, $p_m^{UP}=(1/2)(r+k/2)$, $a_m^{UP}\equiv 1$; for $\min\{3k/2,3t\} \le r < \max\{2t+k/2, t+k\}$ and $k>2t$, $p_m^{UP}=r-t$, $a_m^{UP}\equiv (r-t)/k$; and for $r \ge \max\{t+k, 2t+k/2\}$, $p_m^{UP}=r-t$, $a_m^{UP}\equiv 1$. (Here, subscript m denotes "monopolist," and superscript UP denotes "uniform pricing.") When the optimal price is $2r/3$ or $(1/2)(r+k/2)$, the monopolist does not serve every segment; in this case, the benefit of targeting advertising spending appears as a cost saving—not wasting money reaching segments that will not buy even if made aware.[20] On the other hand, when the optimal price is $r-t$, the monopolist serves every segment, and now, in the absence of price targeting, targeted advertising spending is toothless. This property, as we will see, recurs in the competitive case.

Under targeted prices, the optimal pricing strategy is $p_m(x)=r-tx$ for $x \in [0,1]$. When targeted prices are accompanied by targeted advertising spending, the monopolist chooses $a(\cdot)$ to maximize

$$\int_0^1 \left(p_m(x)a(x)-(k/2)a(x)^2\right)dx,$$

which requires that $a(x)$ maximize $a(x)p_m(x)-(k/2)a(x)^2$ everywhere. The solution is $a_m^*(x)=\min\{p^m(x)/k,1\}$ (here, the asterisk denotes "full targeting"). When targeted prices are accompanied by untargeted advertising spending, the monopolist chooses a to maximize

$$\int_0^1 (ap^m(x)-(k/2)a^2)dx,$$

and the solution now is $a_m^{UA}=\min\{(r-t/2)/k,1\}$ (here, superscript UA denotes "uniform advertising").

Note that the monopolist, when crafting its advertising spending strategy, behaves as a price taker in all three targeting regimes. What differs is the price taken. Under partial advertising targeting, the monopolist is reacting to $\max\{r-t, \min\{2r/3,(1/2)(r+k/2)\}\}$; under full targeting, it is reacting to $r-tx$; and under partial price targeting, it is reacting to $r-t/2$. Comparing the last two, $a_m^{UA}=a_m^*(1/2)$ (i.e., the monopolist's optimal awareness under partial price targeting is the same as the monopolist's optimal awareness *at* 1/2 under full targeting). When $k \le r-t$, $a_m^{UA}\equiv a_m^*(x)\equiv a_m^{UP}(x)\equiv 1$; in this case, the optimal targeted and untargeted awareness levels coincide everywhere in all three solutions. However, in general, given price targeting, optimal targeted awareness is greater than a_m^{UA} in segments closer than 1/2 (where prices are higher than at the median segment), whereas it is lower than a_m^{UA} in segments farther away than 1/2 (where prices are lower than at the median segment). If $k>r$, $a^*(x)<1$ for all x, and in this case, the greater awareness in the stronger half is exactly balanced by the lower awareness in the weaker half, making aggregate awareness the same in both solutions (however, advertising cost is higher in the targeted solution).

Needless to say, full targeting is optimal for a monopolist. With respect to partial price targeting, except when $k \le r-t$ when it is a wash, the benefit of full targeting is that it enables the monopolist to coordinate local awareness levels with local prices—as the Dorfman and Steiner (1954) theorem prescribes. With respect to partial advertising targeting, the benefit is mainly on the pricing side; the monopolist is able to charge every segment what it is willing to pay and as well, serve every segment. However, there is a benefit on the advertising side too; in the presence of price targeting, advertising spending targeting achieves its full potential.

Proceeding to the competitive context, our methodology will be to examine the advertising spending and price equilibrium in the three targeting regimes, one by one, starting with partial advertising targeting. Once we have the three equilibria in hand, we will turn to the question of optimal targeting strategy in Section 7.

3. When Advertising Spending Is Targeted But Price Is Not

This is the simplest case to consider—a straightforward extension of the no targeting frameworks of Grossman and Shapiro (1984) and Tirole (1988, p. 293). Their models restrict both advertising spending and price to be uniform along the Hotelling line; we will allow advertising spending to vary by x. We will call the resulting equilibrium "partial advertising targeting equilibrium." All quantities associated with this equilibrium will bear the superscript "*UP*" to indicate "uniform pricing."

As noted in Section 1, the entire paper of Brahim et al. (2011) is devoted to the partial advertising targeting equilibrium. Instead of regurgitating their analysis, we will focus on conveying the main intuitions. First, why should advertising spending vary by x when price is not varying by x? Absent variation in price, there cannot be a Dorfman–Steiner motivation to vary advertising spending, as we noted in the monopoly analysis. However, in the competitive context, there is another source of variation: the variation in potential market size depending on whether a firm's price is competitive with respect to the "switchers" or not. In other words, at every x, depending on whether $v_j(x) - p_j \geq \max\{v_{3-j}(x) - p_{3-j}, 0\}$ or not, Firm j's ($j = 1, 2$) market at x consists of "loyalists" and "switchers" or just "loyalists."

For $p_2 - t < p_1 < p_2 + t$, the Hotelling line itself divides into two submarkets, $[0, \hat{x}]$ and $[\hat{x}, 1]$, where $\hat{x} = (1/2) + (p_2 - p_1)/2t$. In $[0, \hat{x}]$, the "switchers" prefer Firm 1; in $[\hat{x}, 1]$, the "switchers" prefer Firm 2. Therefore, Firm 1's market share is $a_1(x)(1 - a_2(x)) + a_1(x)a_2(x) = a_1(x)$ at $x \in [0, \hat{x}]$ and $a_1(x)(1 - a_2(x))$ at $x \in [\hat{x}, 1]$. Clearly, there is no reason for Firm 1 to vary $a_1(x)$ within $[0, \hat{x}]$. Similarly, there is no reason for Firm 2 to vary $a_2(x)$ within $[\hat{x}, 1]$. However, if $a_2(x)$ does not vary within $[\hat{x}, 1]$, there is no reason for Firm 1 to vary $a_1(x)$ within $[\hat{x}, 1]$ either. In short, each firm's targeted advertising strategy will have at most two awareness levels, one for the submarket where it is stronger with respect to "switchers," say a_{sj}, and one for the submarket where it is weaker with respect to "switchers," say a_{wj} ($j = 1, 2$).

Hence, the firms' profit functions for $p_2 - t < p_1 < p_2 + t$ will be

$$\pi_1 = p_1(a_{s1}\hat{x} + a_{w1}(1 - a_{s2})(1 - \hat{x})) - \hat{x}(k/2){a_{s1}}^2 - (1 - \hat{x})(k/2){a_{w1}}^2 \tag{1}$$

$$\pi_2 = p_2(a_{s2}(1 - \hat{x}) + a_{w2}(1 - a_{s1})\hat{x}) - (1 - \hat{x})(k/2){a_{s2}}^2 - \hat{x}(k/2){a_{w2}}^2. \tag{2}$$

Maximizing these with respect to a_{sj} and a_{wj} for $j = 1, 2$ yields the following optimal advertising spending strategy:

$$a_{sj} = \min\left\{\frac{p_j}{k}, 1\right\} \tag{3}$$

$$a_{wj} = \min\left\{\frac{p_j(1 - a_{s,3-j})}{k}, 1\right\}. \tag{4}$$

Note that given $(a_{s,3-j}, a_{w,3-j})$, Firm j's optimal advertising strategy depends only on its own price. Furthermore, if $p_j > 0$, optimal $a_{wj} > 0$ if and only if $a_{s,3-j} < 1$. For $k > 2t$, Brahim et al. (2011) show that an interior symmetric pure strategy equilibrium $(a_s^{UP}, a_w^{UP}, p^{UP})$, if it exists, is characterized by $0 < a_w^{UP} < a_s^{UP} < 1$, where $a_s^{UP} = p^{UP}/k$, $a_w^{UP} = (p^{UP}/k)(1 - p^{UP}/k)$, and $p^{UP} \in (2t, t + k/2)$ is the smallest positive root of the first-order condition:

$$\left(\frac{p}{k}\right)^3 - 2\left(1 - \frac{t}{k}\right)\left(\frac{p}{k}\right)^2 - 4\left(\frac{t}{k}\right)\left(\frac{p}{k}\right) + 4\left(\frac{t}{k}\right) = 0. \tag{5}$$

Each firm's profit in this putative equilibrium is

$$\left(p^{UP2}/4k\right)\left(1 + (1 - p^{UP}/k)^2\right). \tag{6}$$

Such an equilibrium will exist only if neither firm can make more money pricing at $r - t$ with awarenesses $\min\{1, (r - t)(1 - a_s^{UP})/k\}$ and $\min\{1, (r - t)(1 - a_w^{UP})/k\}$ in its weaker and stronger halves, respectively (relying exclusively on sales to "loyalists" everywhere). This puts an upper bound on r/t. For $k \leq 2t$, a symmetric pure strategy equilibrium exists if and only if $r \leq (3t + k)/2$; in this equilibrium, $p^{UP} = r - t/2$ (which is less than $t + k/2$ except when $r = (3t + k)/2$).

Proposition 1. *When both firms target advertising spending but not prices*

1. *If $k \leq 2t$, a symmetric pure strategy equilibrium in advertising spending strategies exists if and only if $r \leq (3t + k)/2$. In this equilibrium, $a_s^{UP}(x) \equiv 1$, $a_w^{UP}(x) \equiv 0$, and $p^{UP} = r - t/2$. The firms' profits are $(r - t/2 - k/2)(1/2)$.*
2. *If $k > 2t$, a symmetric pure strategy equilibrium in advertising spending strategies exists only if r/t is not too large. In this equilibrium, $a_s^{UP}(x) \equiv p^{UP}/k$, $a_w^{UP}(x) \equiv (p^{UP}/k)(1 - p^{UP}/k)$, and p^{UP} is the smallest positive root of Equation (5). The firms' profits are as in Equation (6).*

According to this proposition, the partial advertising targeting equilibrium takes two different forms depending on whether $k \leq 2t$ or $k > 2t$. In the latter case, the higher marginal cost of advertising yields an interior solution; both firms advertise everywhere, spending more in their own stronghold than in their competitor's, but nowhere does awareness reach saturation levels. This creates a market for each firm in the opponent's stronghold—the consumers who are *only* aware of it. On the other hand, when $k \leq 2t$, the lower marginal cost of advertising induces each firm to "saturation advertise" in its own stronghold, eliminating opportunities for competitor encroachment in those markets.

A central insight coming out of the partial advertising equilibrium is that advertising spending is in the service of pricing. Its role is to reel in the consumers who are potentially "grabbable" based on the firm's price. Thus, a core function of targeted advertising spending, when prices themselves are nontargeted, is to avoid wastage (i.e., not spend money raising awareness in markets where even if consumers were aware, they would not buy the advertiser's product). This is seen most vividly in the corner equilibrium; neither firm advertises in its weaker half because there is no hope of gaining customers there when the stronger firm is advertising to saturation levels. Another core function of targeted advertising spending is to align the spending level to the potential size of the market, spending more where the market is potentially larger. This is illustrated in the interior equilibrium; firms spend more in their stronghold than in their opponent's stronghold because in the former, their market consists of "loyalists" and "switchers," whereas in the latter, their market is limited to "loyalists." A third core function—aligning local advertising spending to local price—is absent here because local prices do not vary. This function will kick in under full targeting.

From Equation (5), it is clear that the interior equilibrium's price is an increasing function of t, the product differentiation parameter. This is also true in the classical (full-information) Hotelling model—where advertising has no role—and in the Grossman and Shapiro (1984) (Tirole 1988, p. 293) version where advertising has a role similar to ours but is untargeted. This makes sense; the economics of untargeted price competition do not change merely because advertising is present and its spending is targeted. Note, however, that the corner equilibrium's comparative statics go in the opposite way; $r - t/2$ is decreasing in t. This, too, is expected; in this equilibrium, each firm is behaving as a monopolist in its stronghold, and for a monopolist, t is simply a parameter that decreases willingness to pay of the marginal consumer.

Now, we turn to the two targeting regimes involving targeted price competition: targeting price only and full targeting. Each x now behaves as a distinct market; both firms are fighting over the "switchers" within each x instead of the "switchers" at the marginal x ($\hat{x}$). As we will see, this makes a fundamental difference to the nature of the price equilibrium; instead of being in pure strategies, now it will be in mixed strategies. Advertising spending at x sets the conditions for the local price equilibrium by determining the sizes of the "loyalist" and "switcher" subsegments. It follows that the price equilibrium at x depends only on the firms' awareness levels at x; their awareness at locations $x' \neq x$ is immaterial. In other words, the targeted price equilibrium at x given $(a_1(x), a_2(x))$ is a common building block for both the partial price targeting equilibrium as well as the full targeting equilibrium. So, we develop it first.

4. Targeted Price Competition at *x* for Given Awareness Levels

Fix an x; let $a_1 > 0, a_2 > 0$ denote the firms' awareness levels at x. As before, advertising's probabilistic reach creates three subsegments within x: (1) "Firm 1 loyalists," of size $a_1(1 - a_2)$; (2) "Firm 2 loyalists," of size $a_2(1 - a_1)$; and (3) "switchers," of size $a_1 a_2$. Other than this informational differentiation, consumers in x are identical; each of them is willing to pay $v_1(x)$ for Firm 1's product and $v_2(x)$ for Firm 2's product. Because the "loyalists" and "switchers" within x are invisible to the firms, they cannot be explicitly targeted; $p_1(x)$ and $p_2(x)$ will be available to all consumers at x who are aware of Firms 1 and 2, respectively, regardless of whether they are "loyalists" or "switchers."[21]

The following proposition summarizes the resulting price equilibrium at x. Because Firm 1's (Firm 2's) equilibrium strategy at x is the same as Firm 2's (Firm 1's) equilibrium strategy at $1 - x$, it is efficient to state the result in normalized form using stronger firm/weaker firm terminology.

Proposition 2. *The targeted price equilibrium at x, for given $a_s > 0, a_w > 0$, is unique and has the following properties.*

1. *If $a_s < 1$ or $a_w < 1$, the equilibrium is in mixed strategies with the following properties.*

   a. *Firm s's expected revenues are $\underline{p}_s(x)a_s$, and Firm w's expected revenues are $\underline{p}_w(x)a_w$, where $\underline{p}_s(x)$ and $\underline{p}_w(x)$ are the lowest prices in the (closures of the) supports of these firms' price distributions.*

   b. $\underline{p}_s(x) = \underline{p}_w(x) + \Delta(x)$.

   c. *If $a_s/a_w \geq v_s(x)/v_w(x)$, then $\underline{p}_s(x) = v_s(x)(1 - a_w)$, $\underline{p}_w(x) = v_w(x) - v_s(x)a_w$, and the firms' equilibrium price distributions are*

$$F_s(p; x) = \frac{p - v_s(x)(1 - a_w)}{a_s(p - \Delta(x))}, \qquad for \quad p \in [v_s(x)(1 - a_w), v_s(x))$$

$$F_w(p; x) = \frac{p - v_w(x) + v_s(x)a_w}{a_w(p + \Delta(x))}, \qquad for \quad p \in [v_w(x) - v_s(x)a_w, v_w(x)].$$

Firm s has a mass point at $v_s(x)$ of size $(a_s v_w(x) - v_s(x) a_w)/a_s v_w(x)$.

   d. *If $a_s/a_w \leq v_s(x)/v_w(x)$, then $\underline{p}_w = v_w(x)(1 - a_s)$, $\underline{p}_s = v_s(x) - v_w(x)a_s$, and the firms' equilibrium price distributions are*

$$F_s(p; x) = \frac{p - v_s(x) + v_w(x)a_s}{a_s(p - \Delta(x))}, \qquad for \quad p \in [v_s(x) - v_w(x)a_s, v_s(x)]$$

$$F_w(p; x) = \frac{p - v_w(x)(1 - a_s)}{a_w(p + \Delta(x))}, \qquad for \quad p \in [v_w(x)(1 - a_s), v_w(x)).$$

Firm w has a mass point at $v_w(x)$ of size $(a_w v_s(x) - v_w(x) a_s)/a_w v_s(x)$.

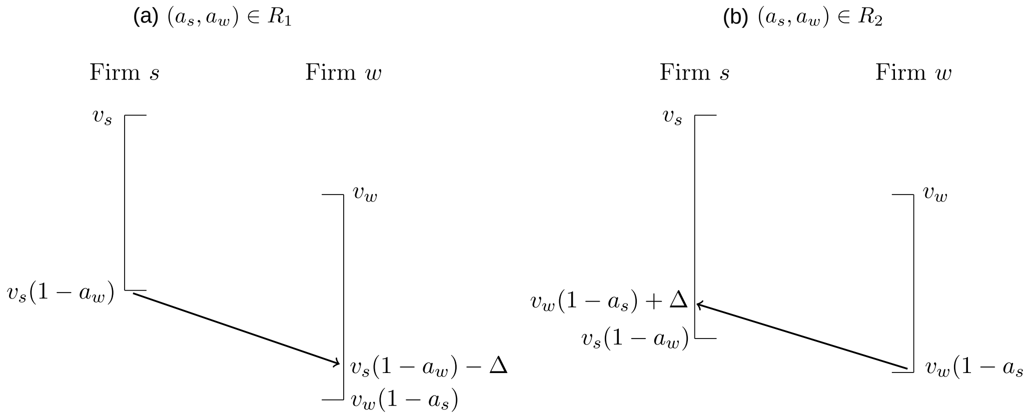
2. *If $a_s = a_w = 1$, then the equilibrium is in pure strategies: $p_s(x) = \Delta(x)$, $p_w(x) = 0$. In this equilibrium, Firm s's revenues are $\Delta(x)$, and Firm w's revenues are zero.*

Proposition 2 states that except when $a_s = a_w = 1$, the local targeted price equilibrium is in mixed strategies, and this equilibrium comes in two varieties depending on whether $a_s/a_w > v_s/v_w$ or $a_s/a_w < v_s/v_w$ (the two varieties blend into each other when $a_s/a_w = v_s/v_w$). For $x \in [0, 1]$, define $R_1(x)$ and $R_2(x)$ as

$$R_1(x) := \{(a_s, a_w) \in [0,1]^2 : a_s/a_w \geq v_s(x)/v_w(x)\}$$
$$R_2(x) := \{(a_s, a_w) \in [0,1]^2 : a_s/a_w \leq v_s(x)/v_w(x)\}.$$

Figure 2 illustrates the construction (without the x notation). Here, $v_s(1 - a_w)$ and $v_w(1 - a_s)$ are, respectively, the prices below which the stronger and weaker firm would not want to go—these being the prices at which

Figure 2. Illustrating the Local Mixed-Strategy Equilibrium in Prices When $a_s < 1$ or $a_w < 1$



Notes. (a) R_1 type. (b) R_2 type.

even capturing all the "switchers" yields the same revenue as getting only "loyalists" at their reservation price. When $a_s/a_w > v_s/v_w$ (i.e., the firms' awareness levels are in R_1), the stronger firm has more awareness than is warranted by its valuation, which by increasing its "loyalist" base, handicaps it in its ability to attract "switchers"; the weaker firm, by contrast, endowed with a smaller "loyalist" base can afford to price more aggressively. This is reflected in panel (a) of Figure 2, where the weaker firm's absolute lower bound, $v_w(1-a_s)$, is lower than $v_s(1-a_w)-\Delta$, the price below which it can capture all the "switchers." In this case, then the stronger firm derives its revenue solely from its "loyalists," whereas the weaker firm derives revenues from both its "loyalists" and "switchers"; this is as if the stronger firm is *implicitly* targeting its "loyalists," whereas the weaker firm is implicitly targeting "switchers."[22]

When $a_s/a_w < v_s/v_w$ (i.e., in R_2), there is a role reversal (see panel (b) of Figure 2); now, the weaker firm is implicitly targeting its "loyalists," whereas the stronger firm is implicitly targeting "switchers."

In short, as long as both firms are not seeking saturation-level awareness in a segment, two types of informational differentiation can arise within a segment, and these play a crucial role in shaping the targeted price equilibrium in mixed strategies that arises at this segment.[23] We will refer to these two types of informational differentiation as R_1- and R_2-type informational differentiation.

Definition 1 (R_1- and R_2-Type Informational Differentiation). If $(a_s(x, a_w(x)) \in R_1(x)$, then we have R_1-type informational differentiation at x; otherwise, we have R_2-type informational differentiation at x.

Proposition 2 will play a crucial role in the full and partial price targeting equilibria to follow. In constructing these equilibria, we will rely on the fact that properties 1, (a) and (b) of this proposition are not simply properties of the local price equilibrium; they are properties of the local price best response functions. In other words, given $a_s(x) > 0, a_w(x) > 0$ with at least one of them greater than one, for *any* price distribution $F_w(\cdot)$ at x with lowest price $\underline{p}_w$—not necessarily the equilibrium $F_w(\cdot; x)$ identified in part 1(c) or part 1(d)—if $F_s(\cdot)$ is Firm s's best price response at x, with lowest price $\underline{p}_s(x)$, it must satisfy the following. (a) Firm s's maximum expected revenues will be $\underline{p}_s(x)a_s(x)$, and (b) $\underline{p}_s(x)$ will equal $\underline{p}_w + \Delta(x)$. Property (a) arises because at the lowest price in the support of a best response price distribution, (i) a firm must maximize its expected revenue, and (ii) it must get all the consumers who are aware of it. Property (b) arises because if Firm s's lowest price is greater or less than $\underline{p}_w + \Delta(x)$, then $F_s(\cdot)$ will be strictly dominated by some other distribution whose lowest price equals $\underline{p}_w + \Delta(x)$. This is similar for Firm w's best price response function.

5. When Both Advertising Spending and Price Are Targeted

When both advertising spending and price are targeted by both firms, we get the full targeting equilibrium; we designate this equilibrium by an $*$ superscript. In this equilibrium, each firm's strategy is a pair of functions $(a_j^*(\cdot), p_j^*(\cdot)), j = 1, 2$, such that

$$(a_j^*(\cdot), p_j^*(\cdot)) \in \arg\max_{(a_j(\cdot), p_j(\cdot))} \int_0^1 \pi_j(a_j(x), p_j(x); a_{3-j}^*(x), p_{3-j}^*(x); x)dx,$$

where $\pi_j(a_j(x), p_j(x); a_{3-j}^*(x), p_{3-j}^*(x); x)$ is Firm j's local profit at x when it plays $(a_j(x), p_j(x))$, whereas Firm $3-j$ plays $(a_{3-j}^*(x), p_{3-j}^*(x))$. An equilibrium of the overall game must, therefore, also be a local equilibrium at each x. That is, $(a_j^*(x), p_j^*(x)), j = 1, 2$, must satisfy

$$(a_j^*(x), p_j^*(x)) \in \arg\max_{(a_j, p_j)} \pi_j(a_j, p_j; a_{3-j}^*(x), p_{3-j}^*(x); x)$$

for all $x \in [0, 1]$. Effectively, each x behaves as a separate market over which the two firms are competing.

Fix an x. Given Proposition 2, it is immediate that any full targeting equilibrium at x must feature at least one firm advertising at less than saturation level. For if $a_1^*(x) = a_2^*(x) = 1$, then neither firm has any "loyalists," and per part (2) of Proposition 2, at least one of them must make negative profits—which it can avoid by choosing zero advertising. Once "loyalists" and "switchers" coexist at an x, equilibrium prices must involve mixed strategies, as specified in part (1) of Proposition 2. Let F_j^* (j = 1, 2) denote Firm j's distribution function in the mixed-strategy equilibrium and $\underline{p}_j^*$ the lowest price in (the closure of) its support.

Turning to advertising strategies, note that each firm takes (a_{3-j}^*, F_{3-j}^*) as given when choosing (a_j, F_j). Once F_{3-j}^* is fixed, $\underline{p}_{3-j}^*$ is fixed. Then, if F_j is part of Firm j's best response, its lowest price, $\underline{p}_j$, is fixed too, *independent of* a_j, being either $\underline{p}_{3-j}^* + \Delta$ (if j is the stronger firm) or $\underline{p}_{3-j}^* - \Delta$ (if j is the weaker firm). In other words, given F_{3-j}^*, Firm j's maximum expected revenue per aware customer is fixed at $\underline{p}_j^*$. It follows then that when choosing advertising spending, each firm will behave as a *price-taking* monopolist, treating $\underline{p}_j^*$ as a parameter while maximizing $\underline{p}_j^* a_j - (k/2)a_j^2$ with respect to a_j.[24]

Define

$$a_w^*(x) := \frac{v_w(x)}{v_s(x) + k},$$
$$a_s^*(x) := \min\left\{ \left(\frac{v_s(x)}{k}\right)(1 - a_w^*(x)), 1 \right\}, \tag{7}$$
$$\underline{p}_w^*(x) := v_w(x) - v_s(x)a_w^*(x) \quad \underline{p}_s^*(x) := v_s(x)(1 - a_w^*(x)) \tag{8}$$

$$F_w^*(p;x) := \frac{p - v_w(x) + v_s(x)a_w^*(x)}{a_w^*(x)(p + \Delta(x))} \qquad \text{for } p \in [\underline{p}_w^*, v_w(x)]$$

$$F_s^*(p;x) := \begin{cases} \dfrac{p - v_s(x)(1 - a_w^*(x))}{a_s^*(x)(p - \Delta(x))} & \text{for } p \in [\underline{p}_s^*, v_s(x)), \\ 1 & \text{for } p = v_s(x) \end{cases}$$

$$\pi_s^*(x) := \underline{p}_s^*(x)a_s^*(x) - (k/2)a_s^*(x)^2$$
$$\pi_w^*(x) := \underline{p}_w^*(x)a_w^*(x) - (k/2)a_w^*(x)^2$$
$$\pi_1^* = \pi_2^* = \pi^* := \int_0^{1/2} (\pi_s^*(x) + \pi_w^*(x))dx. \tag{9}$$

Note that $F_s^*(p;x)$ has a mass point of size $(a_s^*(x)v_w(x) - v_s(x)a_w^*(x))/a_s^*(x)v_w(x)$ at $v_s(x)$ for $x \neq 1/2$. This mass point disappears at $x = 1/2$, and $F_s^*(p;1/2) \equiv F_w^*(1/2) := F^*(p;1/2)$. Furthermore, $a_s^*(x) > a_w^*(x)$ and $\pi_s^*(x) > \pi_w^*(x)$ for $x \neq 1/2$, and $a_s^*(1/2) = a_w^*(1/2) := a^*(1/2)$, $\pi_s^*(1/2) = \pi_w^*(1/2)$.

Proposition 3. *When both firms target advertising spending and prices, there exists a unique equilibrium. In this equilibrium, each firm plays* $(a_s^*(x), F_s^*(x))$ *where it is stronger,* $(a_w^*(x), F_w^*(x))$ *where it is weaker, and* $(a^*(1/2), F^*(1/2))$ *at* $x = 1/2$. *Furthermore,*

1. $a_s^*(x) > a_w^*(x)$, $\underline{p}_s^*(x) > \underline{p}_w^*(x) > 0$, *and* $\pi_s^*(x) > \pi_w^*(x) > 0$ *for* $x \neq 1/2$.
2. R_1*-type informational differentiation prevails at all* x.
3. $\underline{p}_w^*(x)/a_w^*(x) \equiv k$ *and* $\underline{p}_s^*(x)/a_s^*(x) \equiv k$ *if* $a_s^*(x) < 1$.

The equilibrium described in Proposition 3 has several appealing properties. The firms play asymmetric strategies at asymmetric locations and symmetric strategies at the only symmetric location in the Hotelling model—1/2. Furthermore, the asymmetries and symmetries are intuitive. At x = 1/2 where the firms are equally strong, they make the same profit; at $x \neq 1/2$, the stronger firm advertises more, charges a higher (lowest) price, and makes more money, and because the firms are overall symmetric, they make the same total profit. Note that each firm makes positive profit at every location—even locations where it is weaker. As we will see, this property is not guaranteed to hold in the partial price targeting equilibrium.

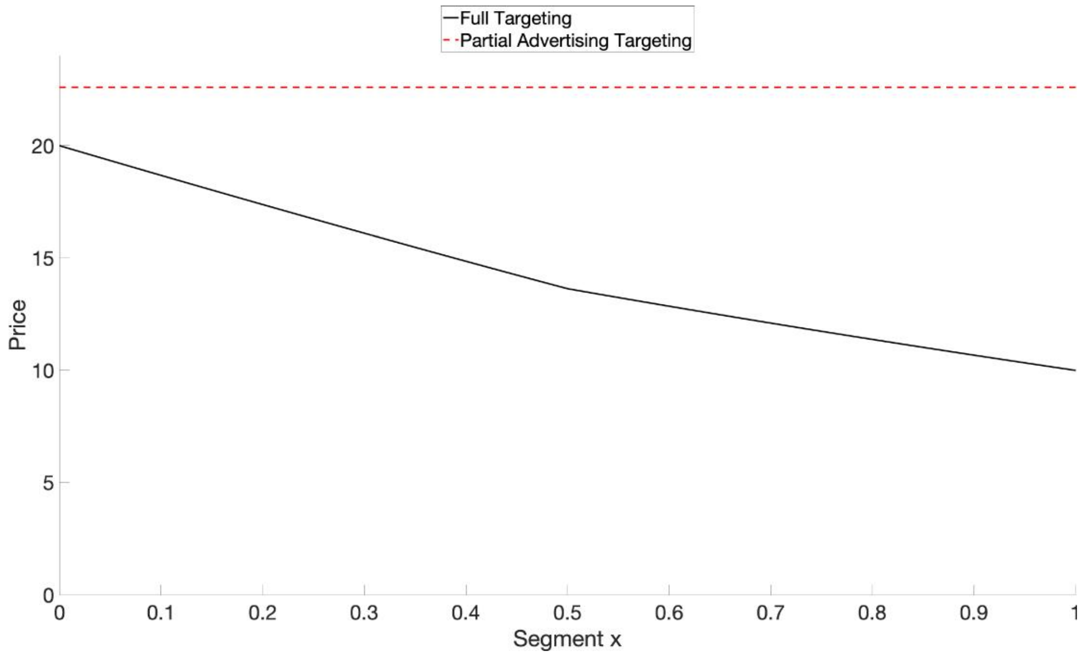
From Equation (8), $\underline{p}_s^*$ increases and $\underline{p}_w^*$ decreases toward the extremities. In other words, each firm increases its lowest price as its competitive advantage (disadvantage) increases (decreases). From Equation (7), the firms' equilibrium awareness levels have the same properties: more awareness at locations where the competitive advantage (disadvantage) is larger (smaller)—as long as it is possible to do so. The weaker firm's lowest price-to-awareness ratio as well as the stronger firm's lowest price-to-awareness ratio in the interior solution are the same as in the monopolist's optimal targeting strategy and thus, satisfy the Dorfman–Steiner theorem. This makes sense given that each firm is behaving as a price-taking monopolist.

Note that $a_w^*(x) < 1$ for all x, but $a_s^*(x) = 1$ when $\sqrt{v_s(x)\Delta(x)} \geq k$. For those x, then the stronger firm chooses not to customize its advertising even though it is allowed to. However, no matter how small k is, $a_s^*(1/2) < 1$, and by continuity, $a_s^*(x) < 1$ also in a neighborhood of 1/2. In other words, even if the marginal cost of generating awareness is negligible, no firm wants saturation awareness throughout the Hotelling line. This also means that both firms will customize awareness over more than half of the Hotelling line given the opportunity to do so.

5.1. Comparing Full vs. Partial Advertising Targeting Equilibrium Strategies

Figure 3 illustrates, from Firm 1's perspective, how equilibrium prices vary along x in the two equilibria. Although prices are constant in the partial advertising targeting equilibrium, by definition, Firm 1's full targeting prices are higher where willingness to pay is higher and competitive advantage is higher. This reflects a fundamental difference between the two targeting regimes.

Figure 3. (Color online) Firm 1's Lowest Price in the Full Targeting Mixed-Strategy Equilibrium Along x Compared with Its Partial Advertising Targeting Equilibrium's Pure Strategy Price ($r = k = 30$, $t = 10$)

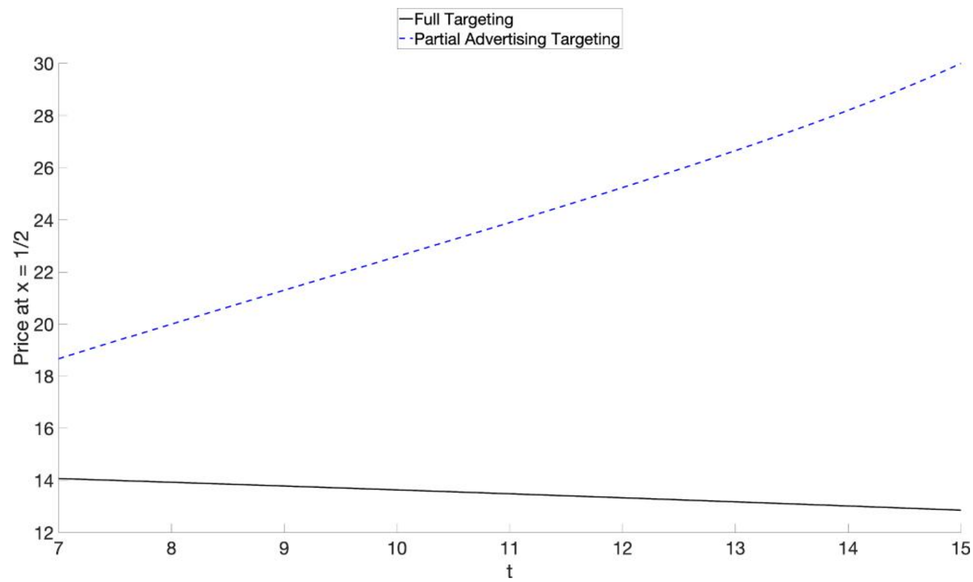


When only advertising spending is targeted, firms compete (in the interior equilibrium) for the "switchers" in the *marginal segment*. The demand function they face is downward sloping, its price elasticity determined by t, the product differentiation parameter; the higher the t, the lower the price elasticity, and hence, the higher the (interior) equilibrium price. By contrast, under full targeting, firms are competing for *each* segment. Because consumers within a segment are homogeneous, this competition is necessarily more cutthroat. The only thing stopping a free fall in prices is the informational differentiation created by advertising's probabilistic reach. Neither firm wants to go very low because each has some "loyalists" who are willing to pay $v_1(x)$ or $v_2(x)$. Also, those valuations are decreasing in t (and in x). Reflecting this intuition, for a given x, $\underline{p}_w^*$—the lowest price of the firm controlling the price equilibrium at that location—is decreasing in t (see (8)). The stronger firm's lowest price at x, however, has a more complicated relationship with t: $\underline{p}_s^*(x) = \underline{p}_w^*(x) + t\,|\,1 - 2x\,|$, where the second term, representing the stronger firm's competitive advantage, is increasing in t. In Figure 4, we take out the second effect by focusing on $x = 1/2$; then, the diametrically opposite effect of t on the two equilibria becomes dramatically evident.

In short, the two equilibria reflect the dual role of t in the Hotelling model. On one hand, it is a parameter that reduces price sensitivity; on the other hand, it is a parameter that reduces willingness to pay. In the partial advertising targeting equilibrium, the first role is operative; in the full targeting equilibrium, the second role is operative. Figure 3 simply reflects this duality. Because t is relatively large here ($r = 30$ and $t = 10$), Firm 1's lowest price in the full targeting equilibrium even at zero is lower than the partial advertising targeting equilibrium price. We should expect this gap to increase as t increases, as shown in Figure 4.

On the advertising front, even though advertising spending is targeted in both cases, it operates quite differently. In the full targeting equilibrium, the function of advertising spending targeting is twofold: (i) determine the terms of the price competition in each segment by setting the relative sizes of the loyalist and switcher subsegments (as discussed) and (ii) allow segment awareness levels to respond to local equilibrium prices, à la Dorfman and Steiner (1954). As Figure 5 shows, the result is awareness levels increasing with competitive advantage. By contrast, in the partial targeting equilibrium, absent price variation, these motivations to vary advertising spending disappear. Instead, the sole motivation to vary advertising spending is variation in market size—a market share multiplier. Market size varies between a firm's stronger and weaker halves because the firm gets "loyalists" and "switchers" in the former but only "loyalists" in the latter. This explains why advertising spending in the partial advertising targeting equilibrium varies only between the two halves—higher in the stronger half—rather than within each half. Indeed, when $k \leq 2t$, awareness in the stronger half is at saturation level, whereas awareness in the weaker half is zero; now, each firm abandons its weaker half because there is no hope of gaining share there.

Figure 4. (Color online) How Equilibrium Prices at $x = 1/2$ Vary with t Under Partial Advertising Targeting vs. Full Targeting ($r = k = 30$)



Note. For full targeting, we are showing Firm 1's lowest price in the mixed-strategy equilibrium at $x = 1/2$.

6. When Price Is Targeted But Advertising Spending Is Not

When price is targeted but advertising spending is not, we get the partial price targeting equilibrium, which we denote with the superscript "*UA*" (indicating "uniform advertising spending").

Define

$$\check{v} := r - \left(\frac{3}{4}\right)t \tag{10}$$

$$a^{UA} := \min\left\{\frac{v}{\check{v} + k}, 1\right\} \tag{11}$$

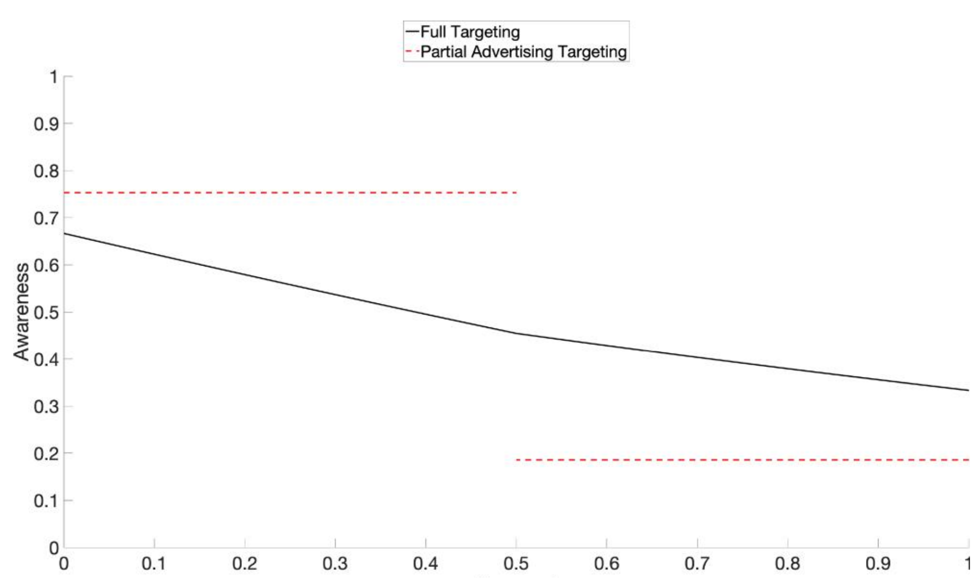
$$\underline{p}_w^{UA}(x) := v_w(x)(1 - a^{UA})$$

$$\underline{p}_s^{UA}(x) := v_s(x) - v_w(x)a^{UA} \tag{12}$$

$$F_w^{UA}(p; x) := \begin{cases} \dfrac{p - v_w(x)(1 - a^{UA})}{a^{UA}(p + \Delta(x))} & \text{for } p \in [\underline{p}_w^{UA}(x), v_w(x)), \\ 1 & \text{for } p = v_w(x) \end{cases}$$

$$F_s^{UA}(p; x) := \frac{p - v_s(x) + v_w(x)a^{UA}}{a_s(p - \Delta(x))}, \quad \text{for } p \in [\underline{p}_s^{UA}(x), v_s(x)]$$

Figure 5. (Color online) Firm 1's Awareness Along x in the Full vs. Partial Advertising Targeting Equilibrium ($r = k = 30$, $t = 10$)



$$\pi_w^{UA}(x) := \underline{p}_w^{UA}(x)a^{UA} - (k/2)a^{UA2}$$
$$\pi_s^{UA}(x) := \underline{p}_s^{UA}(x)a^{UA} - (k/2)a^{UA2} \tag{13}$$
$$\pi^{UA} := \begin{cases} \left(\frac{k}{2}\right)\left(\frac{v}{\check{v}+k}\right)^2 & \text{if} \quad v/(\check{v}+k) < 1 \\ \frac{t}{4} - \frac{k}{2} & \text{if} \quad v/(\check{v}+k) = 1. \end{cases} \tag{14}$$

Note that $F_w^{UA}(p;x)$ has a mass point of size $(v_s(x) - v_w(x))/v_s(x)$ at $v_w(x)$ for $x \neq 1/2$. This mass point disappears at $x = 1/2$, and $F^{UA}(p;1/2) := F_s^{UA}(p;1/2) \equiv F_w^{UA}(p;1/2)$.

Proposition 4. *When both firms target prices but not advertising spending, there exists a unique equilibrium. In this equilibrium, each firm plays $(a^{UA}, F_s^{UA}(x))$ where it is stronger, $(a^{UA}, F_w^{UA}(x))$ where it is weaker, and $(a^{UA}, F^{UA}(1/2))$ at $x = 1/2$. Furthermore,*

1. *$\underline{p}_s^{UA}(x) > \underline{p}_w^{UA}$ and $\pi_s^{UA}(x) > \pi_w^{UA}(x)$ for $x \neq 1/2$.*
2. *$a^{UA} = 1$ if and only if $k \leq t/4$. In this case, there is no informational differentiation anywhere and $\pi_w^{UA}(x) \equiv 0$.*
3. *If $k > t/4$, then R_2-type informational differentiation prevails everywhere, $\underline{p}_w^{UA}(x)/a^{UA} < k$, and $\underline{p}_s^{UA}(x)/a^{UA}$ ranges from $< k$ at $1/2$ to $> k$ at zero and one.*

6.1. Comparing Full vs. Partial Price Targeting Equilibrium Strategies

Examining Equations (7) and (11), it is easy to see that $a^{UA} > a_w^*(x)$ for all $x \in [0,1]$. Furthermore, $a^{UA} > a_s^*(1/2)$, $a_s^*(0) = a_s^*(1) > a^{UA}$ if $a^{UA} < 1$, and $a_s^*(0) = a_s^*(1) = a^{UA}$ if $a^{UA} = 1$. In other words, neither firm uses a^{UA} as its targeted strategy at weaker locations, but both firms use it in a single strong location when $a^{UA} < 1$ and in a positive interval of strong locations when $a^{UA} = 1$. Figure 6 illustrates the former case.

One difference between the full targeting equilibrium and the partial price targeting equilibrium is that although there will be informational differentiation at all locations in the former (because $a_w^*(x) < 1$ at all x), it is possible for the latter to have no informational differentiation anywhere (when $a^{UA} = 1$). Second, to the extent there is informational differentiation in the latter, it will be of type R_2. This is because symmetry coupled with uniform awareness implies $a_s^{UA}(x)/a_w^{UA}(x) \equiv 1 < v_s(x)/v_w(x)$ for $x \neq 1/2$. On the other hand, in the full targeting equilibrium, R_1-type informational differentiation prevails everywhere.

This difference in the type of informational differentiation translates to different implicit targeting strategies in the price equilibrium. Under full targeting, firms are targeting "loyalists" in their stronger segments and "switchers" in their weaker segments; under partial price targeting, it is just the opposite. Which prices are higher depends on x and parameters. At any x, $\underline{p}_s^*(x) - \underline{p}_w^*(x) \equiv \Delta(x) \equiv \underline{p}_s^{UA}(x) - \underline{p}_w^{UA}(x)$, which implies $\underline{p}_s^*(x) > \underline{p}_s^{UA}(x)$ if and only if $\underline{p}_w^*(x) > \underline{p}_w^{UA}(x)$. Now, $\underline{p}_s^*(x) > \underline{p}_s^{UA}(x)$ (and $\underline{p}_w^*(x) > \underline{p}_w^{UA}(x)$) if and only if $v_s(x) > 4(0.5 - x)k$. Therefore, both $\underline{p}_s^*(x) > \underline{p}_s^{UA}(x)$ and $\underline{p}_w^*(x) > \underline{p}_w^{UA}(x)$ in a neighborhood of $1/2$. Elsewhere, the ordering depends on whether $t \geq k$ or not. If $t \geq k$, then $r > 2k$, and $\underline{p}_s^*(x) > \underline{p}_s^{UA}(x)$ and $\underline{p}_w^*(x) > \underline{p}_w^{UA}(x)$ everywhere on $[0,1]$.[25] On the other hand, if $t < k$ and $2t < r < 2k$, $\underline{p}_s^*(x) < \underline{p}_s^{UA}(x)$ and $\underline{p}_w^*(x) < \underline{p}_w^{UA}(x)$ in a neighborhood of zero and one, as illustrated in Figure 7.

Figure 6. (Color online) Firm 1's Awareness Along x in the Full vs. Partial Price Targeting Equilibria ($r = k = 30$, $t = 10$)

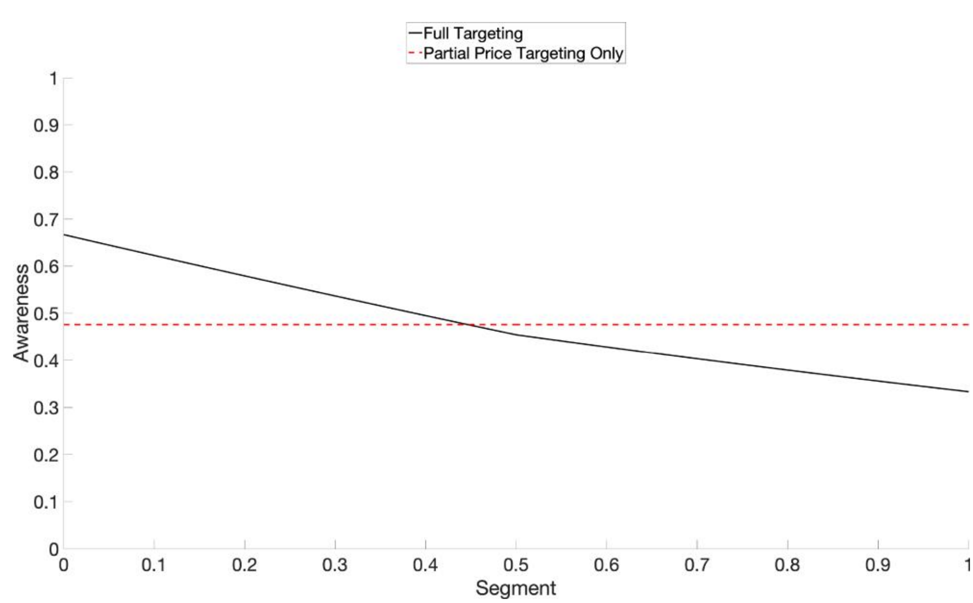
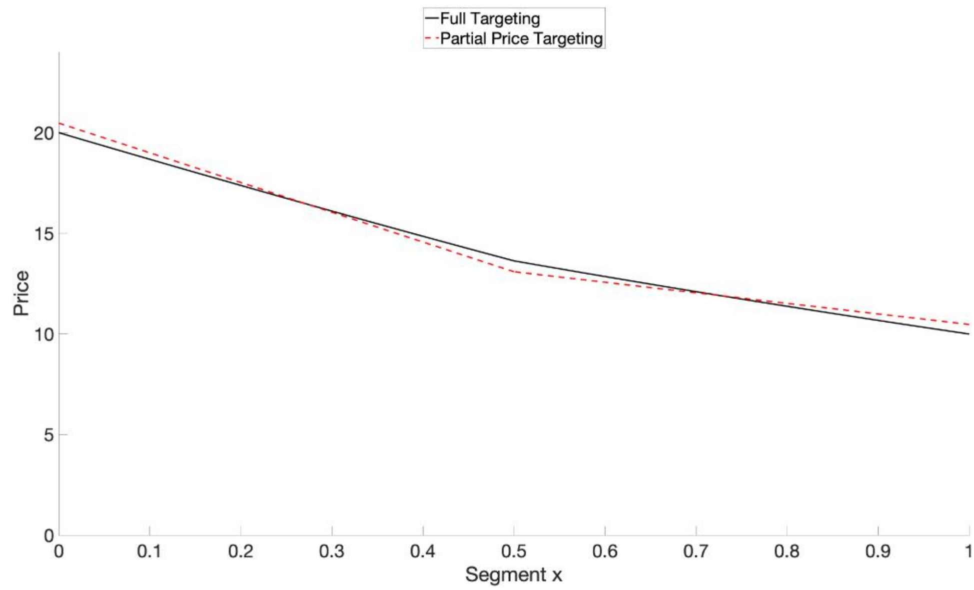


Figure 7. (Color online) Firm 1's Lowest Price in the Full vs. Partial Price Targeting Equilibria as a Function of x ($r = k = 30$, $t = 10$)



7. Should Firms Expand the Scope of Targeting?

Obviously, this question is only relevant if firms are able to choose the scope of their targeting (i.e., they are not constrained to adopt a partial targeting strategy by a third party, such as a retail platform or a regulator). So, let us assume this is the case.

Now, the question can be addressed at two levels: individual rationality and collective rationality. Individual rationality speaks to what the firms would do acting unilaterally in their own self-interest—whether they will target fully or partially and if the latter, whether they will target price only or advertising spending only. We model this as firms choosing among triples—(targeting strategy, advertising spending, price)—where targeting strategy is a choice from the set {partial price targeting, partial advertising targeting, full targeting}, and it is understood that the other two choices respect the constraints, if any, imposed by the targeting strategy choice.

In this framework, is it individually rational for the firms to stay within the targeting constraints embodied in the two partial targeting equilibria? As the proposition asserts, the answer is a categorical no. On the other hand, starting from the full targeting equilibrium, neither firm has an incentive to narrow the scope of its targeting.

Proposition 5. *Absent constraints on the scope of targeting, neither partial targeting equilibrium is sustainable. However, the full targeting equilibrium is.*

The proof of Proposition 5 relies on the fact that, starting from the partial advertising targeting equilibrium of Section 3 or the partial price targeting equilibrium of Section 6, it is individually rational for either firm to expand the scope of targeting. The rationale for expanding the scope of targeting from price targeting only to price and advertising targeting is that, given the competitor's advertising and pricing strategy, it is optimal for each firm to maximize $\underline{p}_j(x)a_j - (k/2)a_j^2$ at each x, treating $\underline{p}_j(x)$ as a parameter. Then, because the parameter varies with x in the partial price targeting equilibrium, optimal awareness levels should vary as well (except when hitting the saturation constraint—and we have already noted that neither firm will hit the saturation constraint everywhere). The rationale for expanding the scope of targeting from advertising spending targeting only to advertising spending and price targeting is even more elemental; because each firm's competitive position varies with x, its best price response will vary with x.

The more interesting question then is the collective rationality question. Are the firms better off in the full targeting equilibrium vis-a-vis either of the partial targeting equilibria? The answer to this question will inform other questions, such as whether firms in an industry should embrace any constraints on the scope of targeting imposed by third parties, whether they should lobby to have them lifted, or whether there is a prisoner's dilemma in the targeting expansion decision.

7.1. Full vs. Partial Price Targeting Equilibrium Profits

Although the firms' profits in the partial targeting equilibrium have a closed-form expression in Equation (9), there is no corresponding expression for the full targeting equilibrium; $\int_0^1 \pi^*(x)dx$ does not have a closed-form expression because the integrands $\pi_s^*(x)$ and $\pi_w^*(x)$ are improper rational functions. However, sufficient conditions for $\pi_s^*(x) > \pi_s^{UA}(x)$ and $\pi_w^*(x) > \pi_w^{UA}(x)$ for all x are easily characterized. Because $\underline{p}_s^*(x) > \underline{p}_s^{UA}(x)$ and $\underline{p}_w^*(x) > \underline{p}_w^{UA}(x)$ if $v_s(x) > 4(0.5 - x)k$, it follows that $k \le t$ implies $\pi_s^*(x), \pi_w^*(x) > \pi^{UA}(x)$ for all $x \in [0,1]$, and hence, $\int_0^1 \pi^*(x)dx > \int_0^1 \pi^{UA}(x)dx$.

Proposition 6. *If $k \le t$, then both firms are better off in the full targeting equilibrium than in the partial price targeting equilibrium.*

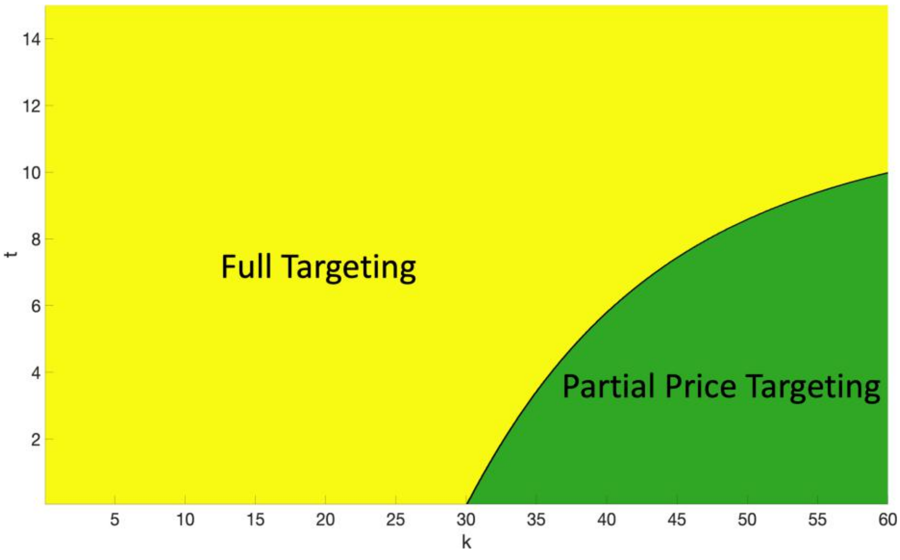
Proposition 6 subsumes $k \le t/4$, and in that case, partial price targeting implies saturation awareness everywhere. The firms' profits are $t/4 - k/2$, which is worse than their profits under no targeting, $t/2 - k/2$ (cf. Tirole 1988, p. 292). In other words, merely introducing advertising into the Hotelling model does not eliminate the prisoner's dilemma effect of targeted pricing. However, as Proposition 6 shows, when price targeting is coupled with advertising spending targeting, then it is possible to avoid the prisoner's dilemma. This is because under full targeting, local awareness can adapt to local prices in accordance with the Dorfman and Steiner (1954) theorem.

It is worth noting that Proposition 6 is conditional on our assumption that the incremental cost of expanding the scope of targeting from price only to price and advertising spending is zero. By continuity, the result will continue to hold if that cost is positive and small enough. However, if that cost is large, Proposition 6 may fail. Whether it does so or not will depend on what the incremental cost of scope expansion actually is, which is an empirical question. Needless to say, we will not have much to say about that except to repeat what we observed in Section 1; to the extent advertising spending targeting and price targeting are based on the same data and the same targeting technology, this cost is likely to be small.

Can partial price targeting ever be better than full targeting? For this to happen, the sufficient condition of Proposition 6 must fail (i.e., k must exceed t). Given $k > t$, $k > r/2$ implies $\underline{p}_s^*(x) < \underline{p}_s^{UA}(x)$ and $\underline{p}_w^*(x) < \underline{p}_w^{UA}(x)$ in a neighborhood of zero and one. If that neighborhood becomes large, both firms may prefer the partial price targeting equilibrium. For example, this is what happens when $r = 30$, $t = 4$, and $k = 50$, as seen in Figure 8, which demarcates the entire (t, k) parameter space under $r = 30$ where the full or partial price targeting equilibrium yields the greater profit. As the figure shows, full targeting is favored for small k and for large k, when t is relatively large; the partial targeting equilibrium is favored elsewhere.

By delegating pricing control to the stronger firm—which has more to lose by pricing lower—partial price targeting reduces the intensity of price competition.[26] This has the effect of raising firms' profits. However, the inability to align advertising spending to profit margins hurts profits. This is most vividly seen when $k \le t/4$; in this case, $a^{UA} = 1$, and neither firm derives profit from its weaker half, even though each is spending $k/4$ advertising dollars generating saturation awareness in that region. In short, adding advertising spending targeting to the targeting mix has costs and benefits for the firms (even in the absence of any out-of-pocket costs). When the costs exceed the benefits, we get a prisoner's dilemma; although individual incentives motivate each firm to expand the scope of targeting, when both do it they end up worse off in the full targeting equilibrium than in the partial price targeting equilibrium.

Figure 8. (Color online) Where Full Targeting Profits Beat Partial Price Targeting Profits and Vice Versa ($r = 30$)



7.2. Full vs. Partial Advertising Targeting Equilibrium Profits

Figure 9 illustrates this comparison. Small t favors full targeting because it reduces product differentiation—which hurts partial advertising targeting profits—while enhancing consumers' willingness to pay for the weaker firm—which improves full targeting profits. The first effect dilutes the main virtue of the partial advertising targeting equilibrium—the softening of price competition under untargeted pricing—whereas the second effect dilutes the main shortcoming of the full targeting equilibrium, namely the delegation of pricing power to the weaker firm. (When the weaker firm's valuation is higher, it is less motivated to price lower.) On the other hand, small k favors the partial advertising targeting equilibrium because it reduces informational differentiation, intensifying price competition in the full targeting equilibrium. At the same time, full targeting's Dorfman–Steiner customization advantage is diluted—the stronger firm's advertising level might hit saturation levels in places, decoupling awareness from prices.

This analysis underscores the central weakness of partial advertising spending targeting vis-a-vis full targeting—its inability to price discriminate between market segments varying in the proportion of "loyalists" to "switchers." This manifests in the firms being unable to compete for "switchers" in their weaker halves, forcing them to rely exclusively on "loyalists" in those markets. However, as noted, partial advertising spending targeting also has a strength; because firms are competing for the "switchers" in the marginal segment, the demand curve they face is downward sloping, which moderates the intensity of price competition.

7.3. Optimal Targeting Strategy

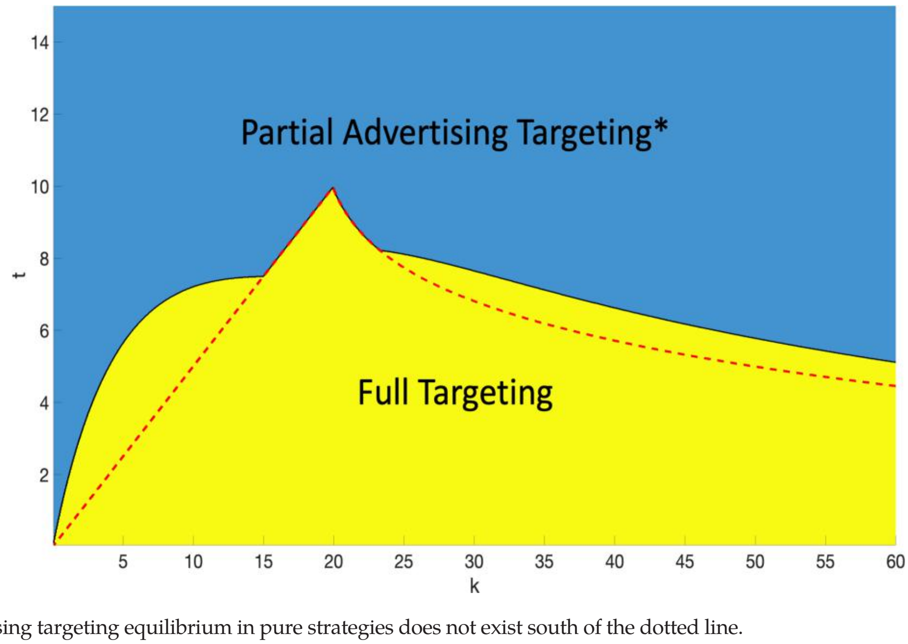
From the foregoing discussion, it should be clear that no targeting regime yields the most profits under all circumstances. Figure 10 identifies when each targeting regime is collectively optimal for $r = 30$. What emerges clearly is that targeting advertising spending only is optimal when t is large (i.e., highly differentiated products operating in high-involvement categories). When products are less differentiated or when product differentiation matters less (as in low-involvement categories), full targeting is optimal when the cost of advertising is small, and partial price targeting is optimal when the cost of advertising is large. We noted in Endnote 4 that online media are among the lowest-CPM advertising vehicles available. The widespread use of advertising spending *and* price targeting in online media such as mobile or email is thus consistent with our results.

8. Discussion

Our analysis shows that expanding the scope of targeting from advertising spending only or price only to both advertising spending and price has benefits and costs for the firms arising from the synergies and dissynergies between the two forms of targeting.

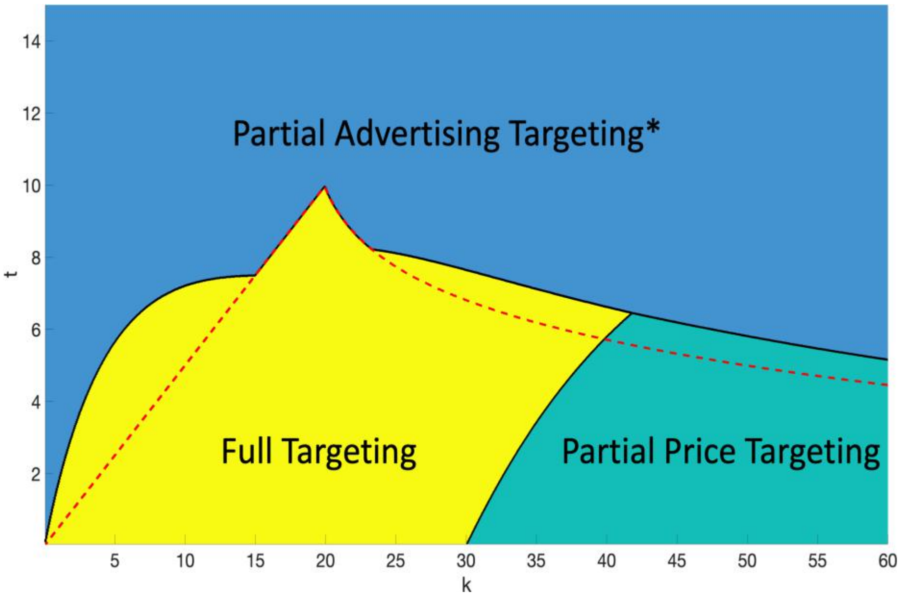
Comparing the full targeting equilibrium with the partial price targeting equilibrium, the fact that both advertising spending and price are targeted in the former allows firms to modulate their local awareness levels in accordance with their local prices (i.e., follow the dictates of the Dorfman–Steiner theorem), just as a monopolist would. This may seem surprising given that our firms are competitors, not monopolists. However, one of the central insights of this paper is that when

Figure 9. (Color online) Where Full Targeting Profits Beat Partial Advertising Targeting Profits and Vice Versa ($r = 30$)



Note. *Partial advertising targeting equilibrium in pure strategies does not exist south of the dotted line.

Figure 10. (Color online) Optimal Targeting Strategy ($r = 30$)



Note. *Partial advertising targeting equilibrium in pure strategies does not exist south of the dotted line.

advertising is informative, price does nearly all the work of competing; advertising's role is to be supportive—making it possible for the firm's price to realize its competitive ambitions. That is why, throughout the paper, each firm's advertising spending decisions resemble those of a price-taking monopolist. What differs among the three targeting regimes is the price taken. Under partial price targeting, firms are making global advertising spending decisions on the basis of average local prices; under full targeting, firms are making local advertising spending decisions based on local prices.

Comparing the partial advertising targeting equilibrium with the full targeting equilibrium, the addition of price targeting allows firms' local prices to respond to local willingness to pay and the firms' local competitive position. In turn, this enhances the functionality of targeted advertising spending; instead of only responding to variations in market size, now advertising spending can respond to variations in price as well.

However, there are costs as well. Notwithstanding the ability to align advertising spending to local prices, it is still the case that price competition is more intense under full targeting. With respect to partial price targeting, this is because full targeting delegates pricing authority to the weaker firm—the firm that has less to lose from lower prices; with respect to partial advertising spending targeting, this is because full targeting substitutes within-segment price competition, regulated by informational differentiation, for between-segment price competition, regulated by product differentiation.

Given benefits and costs, it is not automatic that more targeting is better than less. Firms may be best off under any of the three targeting regimes—full targeting, partial price targeting, or partial advertising spending targeting—depending on the parameter configuration, as shown in Figure 10.

It should be clear from this discussion that these observations should be robust to alternative specifications of the Hotelling model: for instance, replacing linear transportation costs by quadratic transportation costs as in d'Aspremont et al. (1979) or alternative specifications of the advertising cost function such as the one microfounded by Stegeman (1991) for large markets. Similarly, going from a continuous uniform distribution of ideal points to a discrete uniform distribution of ideal points, such as the two-point distribution of Esteves and Resende (2016), should not make a qualitative difference. Finally, going from a uniform distribution to a nonuniform distribution, although it may alter the price and awareness comparisons among the three equilibria along the Hotelling line (such as in Figures 3–5), the aggregate profit comparisons should retain the properties shown in Figure 10 as long as the market is covered under partial advertising targeting as assumed here (Rhodes and Zhou 2022).

8.1. Comparison with Iyer et al. (2005)

Our results contrast sharply with the results in Iyer et al. (2005), the only other paper we are aware of that speaks to the optimal scope of targeting. They find that the marginal value of price targeting given advertising spending targeting is zero, and the marginal value of advertising spending targeting given price targeting is always positive. Translated to optimal targeting strategy, these observations imply that partial price targeting is never collectively optimal and that partial advertising targeting and full targeting are jointly optimal.[27] Our assessments are, of course, quite different. Why might this be so?

We believe there are two major modeling differences between Iyer et al. (2005) and us, one on the demand side and another on the supply side, which together deliver the different results. First, in this paper, targeting is on consumer preferences, which are distributed continuously on the Hotelling line, whereas in Iyer et al. (2005), targeting is on consumers' loyalty (or lack thereof) to the two firms, which makes for two targetable segments per firm. The closest the former comes to the latter is if we were to reimagine the Hotelling model as a discrete model with three consumer types—consumers at the two end points, least likely to switch, and consumers at the midpoint, most likely to switch—but even then, the correspondence would not be perfect. In a Hotelling model, even consumers least likely to switch will consider both firms if they are aware of them, and they will switch if the price is right. By contrast, the loyalists in the Iyer et al. (2005) model never switch. As a result, although firms compete for every segment in the Hotelling model, in the Iyer et al. (2005) model, there is no competition for the loyalist segments.

A sign of the more muted price competition in Iyer et al. (2005) is the lack of a prisoner's dilemma effect when price is targeted. Firms make the same profit with and without price targeting in every information and targeting condition: full information (the Thisse–Vives framework), untargeted advertising, and targeted advertising. The cutthroat competition for the switchers engendered by price targeting simply refuses to spill over to the loyalists. When price targeting has no bite, only advertising spending targeting can offer benefits, and those benefits come primarily as a cost saving. By contrast, in the Hotelling model, price targeting intensifies price competition, reducing firms' profits, but in the presence of advertising spending targeting, is partially redeemed by the opportunity to align local ad spending to local prices—a broader role than cost savings.

Second, the advertising technology is different in two respects. First, advertising spending is a discrete variable in the Iyer et al. (2005) model, whereas it is a continuous variable in our model. Second, advertising exposure is deterministic in the Iyer et al. (2005) model, whereas it is probabilistic in our model. Firms in the Iyer et al. (2005) model face a binary advertising decision with respect to every targeted segment: whether to make everyone in the segment aware or to make no one aware; in equilibrium, each firm randomizes between the two options. When advertising costs are low, the ability to target advertising spending enables each firm to save on the costs of advertising to switchers and the "other" firm's loyalists, and when advertising costs are high (but not too high), there is the further benefit of being able to generate revenues from its own loyalists. Most important, there is no downside to advertising spending targeting, and there is always an upside—the cost saving. By contrast, in our model, advertising spending equilibria exist in pure strategies, and adding advertising spending targeting to the targeting mix has benefits and costs for the firms.

9. Conclusion

In this paper, we have examined the targeting decisions of competing firms on the Hotelling line and asked whether they would benefit from expanding the scope of targeting from advertising spending or price targeting to advertising spending *and* price targeting. The answer is yes and no. Yes, each firm would be individually better off targeting both variables, but no, they *might* end up worse off in the resulting full targeting equilibrium. We identify the conditions under which the full targeting equilibrium or either of the partial targeting equilibria is collectively optimal.

By showing that firms may end up worse off pursuing fuller targeting, this paper may serve as an antidote to popular discussions in the media, which typically see only benefits in the practice. This may be because these discussions are implicitly channeling a monopoly framework, where indeed, more targeting is better than less. In a competitive context, however, expanding the scope of targeting has benefits and costs for the firms arising from the synergies and dissynergies between the two forms of targeting. On the benefit side, going from partial targeting to full targeting allows firms to price discriminate as well as align their local ad spending to local prices—as the Dorfman and Steiner (1954) theorem says they should. On the cost side, going from partial targeting to full targeting increases the intensity of price competition: when price targeting is added, by replacing product differentiation with informational differentiation, and when advertising spending targeting is added, by delegating control of the price equilibrium to the weaker firm (which has a greater incentive to price aggressively). Targeting advertising spending only is collectively optimal when products are highly differentiated, but when product differentiation is low, either full or partial price targeting can be optimal depending on whether the marginal cost of raising awareness is small or large. These results have obvious implications for the design of targeting strategies in various media. For example, retailers geotargeting consumers on mobile devices—a low-cost advertising medium—could benefit from expanding their targeting portfolio from price only to price and advertising spending. For these firms, our paper provides specific guidance on how to design their advertising spending and pricing strategies. At a more macro level, our results speak to the current debate about privacy and targeting practices (Bruell 2021, Haggin and Vranica 2021). Although we have not examined the consumer side of this debate, our results suggest that firms' interests need not be automatically opposed to all targeting restrictions.

Informative advertising may seem like a limited category in which to see these insights, but as we have discussed, this category may actually be broader than it seems. Even so-called persuasive advertising may be informative if it triggers memories that are informative. However, this is not to say that all advertising works in this way or that advertising cannot have purely persuasive effects. Targeted persuasive advertising is, we believe, a completely different animal than targeted informative advertising. We would not be surprised if it had different properties.

Acknowledgments
The authors thank the referees and editors of *Marketing Science* for their comments and suggestions, which have significantly improved the paper. Special thanks to Preyas Desai, Donald Ngwe, and Andrew Rhodes for valuable discussions. The authors also benefited from presentations at various conferences and seminars: the Summer Institute for Competitive Strategy, the University of California, Berkeley 2015; INFORMS Marketing Science Conference 2016; University of Texas at Dallas Forms Conference 2016; Esade Barcelona Pedralbes Campus Marketing Camp 2019; Yale University 2020; and Purdue University 2022.

Appendix
Proof of Proposition 1. For $k > 2t$, the putative interior equilibrium is described in the text; here, we deal only with the existence question. Existence depends on neither firm being tempted by the possibility of pricing at $r - t$ with awarenesses $a'_s = \min\{1, (r-t)(1-a_w^{UP})/k\}$ and $a'_w = \min\{1, (r-t)(1-a_s^{UP})/k\}$ in its stronger and weaker halves, respectively, and relying exclusively on "loyalists." Such a deviation would yield a profit of $(1/2)(r-t)(a'_s(1-a_w^{UP}) + a'_w(1-a_s^{UP})) - (k/4)(a_s'^2 + a_w'^2)$. Hence, equilibrium existence requires

$$\left(\frac{p^{UP2}}{4k}\right)\left(1 + \left(1 - \frac{p^{UP}}{k}\right)^2\right) \geq (1/2)(r-t)(a'_s(1-a_w^{UP}) + a'_w(1-a_s^{UP})) - (k/4)(a_s'^2 + a_w'^2),$$

which imposes an upper bound on r/t. In Figures 9 and 10, we evaluate this inequality numerically.

For the $k \leq 2t$ corner equilibrium, first note that for any $(p_2, a_2(\cdot))$ with $a_2(x) \equiv 1$ for $x \geq 1/2$ and $a_2(x) \equiv 0$ for $x < 1/2$, Firm 1's profit from a price $p_1 < p_2$ with $a_1(x) \equiv 1$ for $x \leq (1/2 + (p_2 - p_1)/2t$ and $a_1(x) \equiv 0$ for $x > (1/2 + (p_2 - p_1)/2t$ is given by $(p_1 - k/2)(1/2 + (p_2 - p_1)/2t)$. Maximizing with respect to p_1 yields $(1/2)(p_2 + t + k/2)$ as Firm 1's optimal undercutting price; this price is indeed lower than p_2 if and only if $p_2 > t + k/2$. For $p_2 \leq t + k/2$, then Firm 1's best response does not involve encroaching on Firm 2's half (i.e., it is optimal for Firm 1 to behave as a local monopolist in its half). As a local monopolist, Firm 1 would choose $p_1 = r - t/2$ with advertising reach 1/2. By symmetry, all of this applies to Firm 2 as well.

Suppose $r \leq (3t + k)/2$. The claim is that $p_1 = p_2 = r - t/2$, $a_s(x) \equiv 1$, and $a_w(x) \equiv 0$ is the unique symmetric pure strategy partial advertising targeting equilibrium in this case. Note first that $r \leq (3t + k)/2 \Longleftrightarrow r - t/2 \leq t + k/2$. Therefore, from the discussion, neither firm has a profitable lower-priced deviation. Nor does either firm have a higher-priced deviation because as a monopolist, either firm would like to price at $(1/2)(r + k/2)$, even lower than $r - t/2$. Hence, $p_1 = p_2 = r - t/2$, $a_s(x) \equiv 1$, and $a_w(x) \equiv 0$ is indeed an equilibrium. The equilibrium is unique because if (p, p) is another equilibrium with $p < r - t/2$, then either firm can make more money by deviating to $r - t/2$ while preserving its advertising strategy.

Now, suppose $r > (3t + k)/2$. In this case, $r - t/2 > t + k/2$. If a symmetric equilibrium exists with $a_s(x) \equiv 1$, $a_w(x) \equiv 0$, and $p < r - t/2$, then either firm can make more money by raising its price to $r - t/2$ while keeping is advertising strategy unchanged; if $p = r - t/2$, then either firm can make more money by lowering its price to $(2r + t + k)/4$ while expanding its advertising reach to $(2r + t - k)/8t > 1/2$ (as seen). In short, there cannot be a symmetric pure strategy equilibrium if $r > (3t + k)/2$ and $k \leq 2t$. □

Proof of Proposition 2. Previous treatments of the price equilibrium have already appeared in the literature (Narasimhan 1988, McAfee 1994, Chen and Iyer 2002), so we provide only a sketch of the proof.

Without loss of generality, fix an $x \leq 1/2$. Because Firm 1 is the stronger firm and Firm 2 is the weaker firm, we can replace the s and w subscripts in the proposition by one and two, respectively. If $a_1 = a_2 = 1$ at x, then neither firm has any "loyalists" there, and the price equilibrium must be $p_1 = \Delta, p_2 = 0$ (Thisse and Vives 1988), which is part (2) of the proposition. Now, assume $a_1 < 1$ or $a_2 < 1$ at x. Because at least one firm has "loyalists" now, from familiar arguments (e.g., Narasimhan 1988), there is no equilibrium in pure strategies. Let F_1 and F_2 denote the two firms' cumulative distribution functions in the mixed-strategy equilibrium at x and $\underline{p}_1$ and $\underline{p}_2$ the lowest prices in the (closures of the) supports of those functions. It follows that $\underline{p}_1 = \underline{p}_2 + \Delta$, and because each firm must capture its entire potential market at its lowest price, their equilibrium revenues are $a_1\underline{p}_1$ and $a_2\underline{p}_2$, respectively. For Firm 1, because it can guarantee revenues of $a_1(1 - a_2)v_1$, it must be that $\underline{p}_1 \geq v_1(1 - a_2)$; similarly, for Firm 2, $\underline{p}_2 \geq v_2(1 - a_1)$. If $v_1(1 - a_2) - \Delta > v_2(1 - a_1)$ (i.e., $a_1/a_2 > v_1/v_2$), then Firm 2 controls the "switching" market, and $\underline{p}_1 = v_1(1 - a_2)$, $\underline{p}_2 = v_1(1 - a_1) - \Delta$ (see Figure 2(a)); if $v_2(1 - a_1) + \Delta > v_1(1 - a_2)$ (i.e., $a_1/a_2 < v_1/v_2$), then Firm 1 controls the "switching" market, and $\underline{p}_1 = v_2(1 - a_1) + \Delta$, $\underline{p}_2 = v_2(1 - a_1)$ (see Figure 2(b)).

The equilibrium distribution functions are obtained by noting that each firm's randomization must make the other firm indifferent among all the prices in the support of its distribution. For instance, if $a_1/a_2 \geq v_1/v_2$, F_1 is obtained by solving $p[a_1a_2(1 - F_1(p + \Delta)) + a_2(1 - a_1)] = (v_1(1 - a_2) - \Delta)a_2$ for $p \in [v_1(1 - a_2) - \Delta, v_2]$. The mass point at v_1, if any, is calculated as $1 - F_1(v_1)$. □

Proof of Proposition 3. As discussed in the text, given $(a_{3-j}, F_{3-j}(\cdot))$ (j = 1, 2), if F_j is part of Firm j's best response at x, then it must satisfy $\underline{p}_1 = \underline{p}_2 + \Delta$ if $x \in [0, 1/2]$ and $\underline{p}_2 = \underline{p}_1 + \Delta$ if $x \in (1/2, 1]$, for *any* $a_j > 0$. Therefore, when choosing a_j, Firm j will treat $\underline{p}_j$ as a constant while maximizing $a_j\underline{p}_j - (k/2)a_j^2$. In other words, if $(a_1^*, F_1^*), (a_2^*, F_2^*)$ is a local equilibrium at x, then $p_1^* = \underline{p}_2^* + \Delta$ if $x \in [0, 1/2]$ and $\underline{p}_2^* = \underline{p}_1^* + \Delta$ if $x \in (1/2, 1]$, and a_j^*, $j = 1, 2$, must satisfy the first-order condition

$$\underline{p}_j^* - ka_j^* \geq 0 \qquad \text{(with equality if } a_j^* < 1).$$

Fix an $x \leq 1/2$. First, we will argue that there cannot be a local equilibrium satisfying $a_1^*/a_2^* < v_1/v_2$. Suppose there is. By Proposition 2, $\underline{p}_1^* = v_2(1 - a_1^*) + \Delta = v_1 - v_2 a_1^*$ and $\underline{p}_2^* = v_2(1 - a_1^*)$. Then, the advertising first-order conditions imply $a_2^* = (v_2/k)(1 - a_1^*)$, and $a_1^* = \min\{v_1/(v_2 + k), 1\}$, but these violate $a_1^*/a_2^* < v_1/v_2$.

Hence, a local equilibrium must satisfy $a_1^*/a_2^* \geq v_1/v_2$. We will construct such an equilibrium. By Proposition 2, because Firm 2 controls the price equilibrium, $\underline{p}_2^* = v_1(1 - a_2^*) - \Delta = v_2 - v_1 a_2^*$. Then, $\underline{p}_1^* = \underline{p}_2^* + \Delta = v_1(1 - a_2^*)$. Because $\underline{p}_1^* \geq \underline{p}_2^*$ and at most one of a_1^* and a_2^* can be equal to one, it follows that $a_2^* < 1$. The advertising first-order condition for Firm 2 is thus satisfied as an equality, and we get

$$a_2^* = \frac{v_2}{v_1 + k}. \tag{A.1}$$

Substituting for $\underline{p}_1^*$ in the advertising first-order condition for Firm 1, we get

$$a_1^* = \min\left\{\left(\frac{v_1}{k}\right)(1 - a_2^*), 1\right\}. \tag{A.2}$$

It is easy to verify that $\min\{(v_1/k)(1 - a_2^*), 1\}/(v_2/(v_1 + k)) \geq v_1/v_2$. In other words, we have the unique local equilibrium at x. Simple computations reveal that this equilibrium satisfies parts (1)–(3) of the proposition. □

Proof of Proposition 4. Once again, at any x, given (a_{3-j}, F_{3-j}) ($j = 1, 2$), if F_j is part of Firm j's best response at x, then it must satisfy $\underline{p}_1 = \underline{p}_2 + \Delta$ if $x \in [0, 1/2]$ and $\underline{p}_2 = \underline{p}_1 + \Delta$ if $x \in (1/2, 1]$, for any $a_j > 0$. In other words, when choosing its best untargeted advertising response to $(a_{3-j},\ F_{3-j}(\cdot))$, each firm j will treat $\int_0^1 \underline{p}_j(x)dx$ as a constant while solving $\max_{a_j} a_j \int_0^1 \underline{p}_j(x)dx - (k/2)a_j^2$. In particular, if $(a_1^{UA}, F_1^{UA}(\cdot)), (a_2^{UA}, F_2^{UA}(\cdot))$ is an equilibrium with untargeted advertising strategies, then a_j^{UA} must satisfy the first-order conditions

$$\int_0^1 \underline{p}_j^{UA}(x)dx - ka_j^{UA} \geq 0 \qquad \text{(with equality if } a_j^{UA} < 1)$$

for $j = 1, 2$.

First, we will show that $\int_0^1 \underline{p}_1^{UA}(x)dx \neq \int_0^1 \underline{p}_2^{UA}(x)dx$ is impossible in a partial price targeting equilibrium. Suppose, to the contrary, that it is possible. Without loss of generality, assume $\int_0^1 \underline{p}_1^{UA}(x)dx > \int_0^1 \underline{p}_2^{UA}(x)dx$. Then, the advertising first-order conditions imply $a_1^{UA} \geq a_2^{UA}$, with equality prevailing if and only if $a_1^{UA} = a_2^{UA} = 1$. However, a_1^{UA} and a_2^{UA} cannot both be one, so it must be that $a_2^{UA} < a_1^{UA} \leq 1$.[28] Hence, $(a_1^{UA}, a_2^{UA}) \in R_1(1/2)$ and $(a_1^{UA}, a_2^{UA}) \in R_2(x)$ for $x \in (1/2, 1]$. Furthermore, if $a_1^{UA}/a_2^{UA} \leq v_1(0)/v_2(0)$, there exists a $x^u \in [0, 1/2)$ such that $a_1/a_2 = v_1(x^{UA})/v_2(x^{UA})$, so that $(a_1^{UA}, a_2^{UA}) \in R_1(x)$ for $x \in [x^{UA}, 1/2]$ and $(a_1^{UA}, a_2^{UA}) \in R_2(x)$ for $x \in [0, x^{UA}]$. On the other hand, if $a_1^{UA}/a_2^{UA} > v_1(0)/v_2(0)$, then $x^{UA} < 0$, effectively collapsing the $[0, x^{UA}]$ interval. To summarize, $(a_1^{UA}, a_2^{UA}) \in R_1(x)$ for $x \in [\max\{x^{UA}, 0\}, 1/2]$ and $(a_1^{UA}, a_2^{UA}) \in R_2(x)$ for $x \in [0, \max\{x^{UA}, 0\}] \cup (1/2, 1]$.

From Proposition 2, we then have

$$\underline{p}_1^{UA}(x) = \begin{cases} v_1(x) - v_2(x)a_1^{UA} & \text{for } x \in [0, \max\{x^{UA}, 0\}] \\ v_1(x)(1 - a_2^{UA}) & \text{for } x \in [\max\{x^{UA}, 0\}, 1/2] \\ v_1(x)(1 - a_2^{UA}) & \text{for } x \in (1/2, 1] \end{cases}$$

$$\underline{p}_2^{UA}(x) = \begin{cases} v_2(x)(1 - a_1^{UA}) & \text{for } x \in [0, \max\{x^{UA}, 0\}] \\ v_2(x) - v_1(x)a_2^{UA} & \text{for } x \in [\max\{x^{UA}, 0\}, 1/2] \\ v_2(x) - v_1(x)a_2^{UA} & \text{for } x \in (1/2, 1]. \end{cases}$$

Substituting for $\underline{p}_1^{UA}(x)$ and $\underline{p}_2^{UA}(x)$ in the first-order conditions for the two firms, we get

$$\begin{aligned} &\int_0^{\max\{x^{UA},0\}} (v_1(x) - v_2(x)a_1^{UA})dx \\ &\quad + \int_{\max\{x^{UA},0\}}^1 v_1(x)(1 - a_2^{UA})dx - ka_1^{UA} \\ &= \int_0^1 v_1(x)dx - a_1^{UA}\int_0^{\max\{x^{UA},0\}} v_2(x)dx \\ &\quad - a_2^{UA}\int_{\max\{x^{UA},0\}}^1 v_1(x)dx - ka_1^{UA} \geq 0 \qquad \text{and} \end{aligned}$$

$$\begin{aligned} &\int_0^{\max\{x^{UA},0\}} v_2(x)(1 - a_1^{UA})dx \\ &\quad + \int_{\max\{x^{UA},0\}}^1 (v_2(x) - v_1(x)a_2^{UA})dx - ka_2^{UA} \\ &= \int_0^1 v_2(x)dx - a_1^{UA}\int_0^{\max\{x^{UA},0\}} v_2(x)dx \\ &\quad - a_2^{UA}\int_{\max\{x^{UA},0\}}^1 v_1(x)dx - ka_2^{UA} = 0. \end{aligned}$$

The left-hand sides of these first-order conditions are identical except for $ka_1^{UA} > ka_2^{UA}$; hence, they cannot be satisfied simultaneously. Therefore, there cannot be an equilibrium satisfying $\int_0^1 \underline{p}_1^{UA}(x)dx \neq \int_0^1 \underline{p}_2^{UA}(x)dx$.

Now, we will construct a partial price targeting equilibrium satisfying $\int_0^1 \underline{p}_1^{UA}(x)dx = \int_0^1 \underline{p}_2^{UA}(x)dx$. In this case, the first-order conditions imply $a_1^{UA} = a_2^{UA}$. In turn, $a_1^{UA} = a_2^{UA}$ implies $(a_1^{UA}, a_2^{UA}) \in R_2(x)$ for all $x \in [0, 1]$. Therefore, from Proposition 2, $p_s^{UA}(x) = v_s(x) - v_w(x)a_s^{UA}$ and $\underline{p}_w^{UA}(x) = v_w(x)(1 - a_s^{UA})$ for all $x \in [0, 1]$. Substituting for $\underline{p}_1^{UA}(x)$ in the advertising first-order condition for Firm 1,

$$\begin{aligned} &\int_0^{1/2} (v_1(x) - v_2(x)a_1^{UA})dx + \int_{1/2}^1 v_1(x)(1 - a_2^{UA})dx - ka_1^{UA} \\ &= \int_0^{1/2} ((r - tx) - (r - t(1 - x))a_1^{UA})dx \\ &\quad + \int_{1/2}^1 (r - tx)(1 - a_2^{UA})dx - ka_1^{UA} \\ &= \left(\frac{4r - t}{8}\right) - \left(\frac{4r - 3t}{8}\right)a_1^{UA} + \left(\frac{4r - 3t}{8}\right)(1 - a_2^{UA}) - ka_1^{UA} \\ &\geq 0. \end{aligned}$$

Setting $a_1^{UA} = a_2^{UA}$ and solving for a_1^{UA}, we get

$$a_1^{UA} = a_2^{UA} = a^{UA} = \min\left\{\frac{v}{\check{v}+k}, 1\right\}, \tag{A.3}$$

where $\check{v} = r - (3/4)t$. Note that $a^{UA} = 1$ if and only if $k \le t/4$.

In short, the unique equilibrium is the one given in Equation (A.3) coupled with the price distribution functions

$$F_s^{UA}(p;x) = \frac{p - v_s(x) + v_w(x)a^{UA}(x)}{a_s(p - \Delta(x))}, \quad \text{for } p \in [v_s(x) - v_w(x)a^{UA}, v_s(x)]$$

$$F_w^{UA}(p;x) = \frac{p - v_w(x)(1 - a^{UA})}{a^{UA}(p + \Delta(x))}, \quad \text{for } p \in [v_w(x)(1 - a^{UA}), v_w(x)),$$

in which Firm w has a mass point of size $(v_s(x) - v_w(x))/v_s(x)$ at $v_w(x)$. The two firms' equilibrium profits are

$$\pi_1^{UA} = \pi_2^{UA} = \pi^{UA} = \begin{cases} \left(\frac{k}{2}\right)\left(\frac{v}{\check{v}+k}\right)^2 & \text{if } v/(\check{v}+k) < 1 \\ \frac{t}{4} - \frac{k}{2} & \text{if } v/(\check{v}+k) = 1. \end{cases} \tag{A.4}$$

Straightforward computations reveal that if $k > t/4$, then $\underline{p}_w^{UA}(x)/a^{UA} = (v_w(x)/v)(k - t/4) < k$ and $\underline{p}_s^{UA}(x)/a^{UA} = (v_s(x)/v)(k + \check{v} - (v_w(x)/v_s(x))v) \in [k - t/4, (r/v)(k + (t/2)(1.5 - t/r))]$ for $x \in [0, 1/2]$. Note that $(r/v)(k + (t/2)(1.5 - t/r)) > k$. □

Proof of Proposition 5. Absent constraints on targeting, each firm is choosing a triple (targeting strategy, advertising spending, price), where targeting strategy is a choice from the set {partial price targeting, partial advertising targeting, full targeting}, and it is understood that each firm's advertising spending and price choices must respect the constraints, if any, set by the targeting strategy choice.

Because partially targeted advertising and price strategies are special cases of fully targeted strategies, the full targeting equilibrium examined in Proposition 3 continues to be an equilibrium in the expanded strategy space. What remains is to evaluate whether ((partial price targeting, $a^{UA}, F^{UA}(\cdot)$), (partial price targeting, $a^{UA}, F^{UA}(\cdot)$)), where a^{UA} and $F^{UA}(\cdot)$ are defined as in Proposition 4, or ((partial advertising targeting, $a^{UP}(\cdot)$, p^{UP}), (partial advertising targeting, $a^{UP}(\cdot)$, p^{UP})), where $a^{UP}(\cdot)$ and p^{UP} are defined as in Proposition 1, can also be sustained as equilibria in the expanded strategy space.

Consider first the former. In $[0, 1/2]$, $\underline{p}_1^{UA}(x) = v_1(x) - v_2(x)a^{UA}$ and $\underline{p}_2^{UA}(x) = v_2(x)(1 - a^{UA})$. If $a^{UA} = 1$, then $\pi_2^{UA}(x) \equiv -k/2 < 0$, and Firm 2 can do better by switching to full targeting and choosing $a_2(x) = 0$ in its weaker half while changing nothing in its stronger half; if $a^{UA} < 1$, then Firm 2 can do better in its weaker half by switching to full targeting, playing $\underline{p}_2^{UA}(x)$ as a pure strategy, and changing its advertising strategy to $a_w(x) = \min\{v_w(x)(1 - a^{UA})/k, 1\}$, the optimal targeted advertising spending strategy corresponding to $\underline{p}_2^{UA}(x)$. In short, the partial price targeting equilibrium is not sustainable once the constraints on targeting advertising spending are lifted.

Now, consider the partial advertising targeting equilibrium in Proposition 1. In this equilibrium, if $k \le 2t$ and $r \le (3t + k)/2$, $a_s^{UP}(x) \equiv 1$, $a_w^{UP}(x) \equiv 0$, and $p^{UP} = r - t/2$. Now, each firm can do better in its stronger half by switching to full targeting and raising its price for $x < 1/2$ to $v_s(x)$ while keeping everything else the same. If $k > 2t$, then each firm can do better in its stronger half by switching to full targeting and raising its price at x to $p^{UP} + \Delta(x)$ if $p^{UP} + \Delta(x) > v_s(x)(1 - a^{UP}(x))$ or to $v_s(x)$ if $p^{UP} + \Delta(x) \le v_s(x)(1 - a^{UP}(x))$, while keeping everything else the same. In short, both firms have unilateral incentives to deviate from the partial advertising targeting equilibrium once the constraints on price targeting are lifted. □

Proof of Proposition 6. $\underline{p}_s^*(x) = v_s(x)(1 - a_w^*(x)) > v_s(x) - v_w(x)a^{UA} = \underline{p}_s^{UA}(x)$ and $\underline{p}_w^*(x) = v_w(x) - v_s(x)a_w^*(x) > v_w(x)(1 - a^{UA}) = \underline{p}_w^{UA}(x)$ if and only if $a^{UA} > v_s(x)/(v_s(x) + k)$. If $k \le t/4$, then $a^{UA} = 1$, and this is true for all $x \in [0, 1]$. Then, because $a_s^*(x)$ maximizes $\underline{p}_s^*(x)a - (k/2)a^2$ and $a_w^*(x)$ maximizes $\underline{p}_w^*(x)a - (k/2)a^2$ and those expressions are greater than $\underline{p}_s^{UA}(x)a - (k/2)a^2$ and $\underline{p}_w^{UA}(x)a - (k/2)a^2$, respectively, $\pi_s^*(x) > \pi_s^{UA}(x)$ and $\pi_w^*(x) > \pi_w^{UA}(x)$ for all $x \in [0, 1]$. If $t \ge k > t/4$, then $a^{UA} = v/(\check{v} + k) > v_s(x)/(v_s(x) + k)$ almost everywhere in $[0, 1]$ if and only if $v/(\check{v} + k) \ge r/(r + k)$ (i.e., $k \le r/2$, which is true because $r > 2t$). Again, $\pi_s^*(x) > \pi_s^{UA}(x)$ and $\pi_w^*(x) > \pi_w^{UA}(x)$ almost everywhere, and the result follows. □

Endnotes

[1] There are exceptions, of course, and those exceptions arise when online advertising is combined with offline prices. For example, a firm may want to advertise by different amounts to old and new consumers as well as charge them different prices. However, although the former is easy to implement online, the latter is hard to do in a brick-and-mortar store.

[2] Indeed, the paper by Sahni et all. (2016) is titled "Do targeted discount offers serve as advertising? Evidence from 70 field experiments," and the paper by Dubé et al. (2017) paper notes that a "limitation of our experimental design is that mobile coupons not only reduce the net price paid by a consumer but also have a promotional effect that could also shift demand" (Dubé et al. 2017, p. 951).

[3] The paper by Thisse and Vives (1988) has inspired a large price-targeting literature in the same no role for advertising vein. Some replicate their prisoner's dilemma effects (Shaffer and Zhang 1995), whereas others show that those effects can be overturned (Corts 1998, Shaffer and Zhang 2002). Rhodes and Zhou (2022) provide a general analysis of the conditions under which targeted price competition benefits firms and consumers.

[4] See Solomon Partners (2021) for estimates of the CPM of different advertising media.

[5] Ever since the original geographic interpretation of his model by Hotelling (1929), competition between location-differentiated brick-and-mortar retailers has been the prototypical application of the Hotelling model (even though it applies more broadly to any kind of horizontal product differentiation). In the modern era, targeting by brick-and-mortar retailers takes the form of geotargeted promotions on mobile phones (Fong et al. 2015, Dubé et al. 2017, Chandra 2021). For example, Chandra (2021) notes that Hyundai, Whole Foods Market, and Dunkin Donuts all have active geoconquesting programs—these firms seek to persuade even consumers located at a competitor's brick-and-mortar store to switch to their store. The field experiments of Fong et al. (2015) and Dubé et al. (2017) confirm the relevance of the Hotelling model in these applications.

[6] These "loyalists" and "switchers" are obviously different from the loyalists and switchers in the Iyer et al. (2005) model. Whereas theirs are born that way, ours are created endogenously by advertising's

probabilistic reach—that is why we put them in quotes. Furthermore, although Iyer et al. (2005) loyalists and switchers are assumed to be identifiable and hence, targetable, our "loyalists" and "switchers," being the product of differential exposure to advertising, are invisible to the firms and hence, untargetable.

[7] The targeting is implicit because "loyalists" and "switchers," being invisible to the firms, cannot be targeted explicitly; see Endnote 6.

[8] See also Grossman and Shapiro (1984), Tirole (1988, section 7.3), Iyer et al. (2005), and Christou and Vettas (2008).

[9] The "location" metaphor is well entrenched in the literature following the Hotelling (1929) original geographic interpretation of his model, but as he himself recognized, its applications extend far beyond the geographic realm to encompass horizontal product differentiation generally. As Anderson et al. (1992, p. 9) observe, "[a] point in space is a *label* for a grocery store's location in Podunk (USA), for the Tories' platform in the ideological spectrum, and for the dryness of cider."

[10] This may well be true for rarely purchased goods and services, such as auto body repair and tour operators (Pazgal et al. 2022)

[11] For example, consumers in a shopping mall may receive a geotargeted ad on their mobile phones from one of the stores in the mall, which then prompts them to visit the store. Sharp (2010) suggests that "refreshing memory structures" is, in fact, the primary function of advertising for most brands. Clark et al. (2009) and Lobschat et al. (2017) provide indirect evidence for the rekindling of memories and the updating of information.

[12] In the Iyer et al. (2005) model, as noted earlier, advertising exposure is deterministic. Either all consumers in a targeted segment become aware (if the requisite money is spent) or nobody becomes aware (if the requisite money is not spent).

[13] Probability of exposure and expected fraction exposed are related because for a random sample of size n from a Bernoulli distribution with parameter a, $E(\sum_1^n X_i/n) = an/n = a$.

[14] The quadratic advertising cost function would ordinarily be seen as simply a tractable operationalization of a convex advertising cost function. In our context, however, there is additional motivation to use it; Chen and Iyer (2002) and Brahim et al. (2011) use it, and we would like to be able to compare our equilibria with theirs. As to why advertising costs should be convex in the probability of reaching a random consumer, the most straightforward explanation is that as more consumers are reached, the chances of reaching an already-reached consumer increase (i.e., the chances of reaching a "new" consumer decrease). Stegeman (1991) provides a formal derivation of convexity for a large market. Clark et al. (2009) and Vilcassim et al. (1999) offer empirical support.

[15] Goeree (2008) provides evidence that differential ad exposure affects consideration set size and composition in the PC market. She further notes that not recognizing this dependence will lead to price elasticity estimates that are too high.

[16] The reason "loyalists" and "switchers" are invisible to the firms is because firms cannot observe consumers' ad exposure—see the discussion.

[17] Given that the price discrimination in question is on the basis of reservation prices, if a monopolist was doing it, we would be calling this first-degree price discrimination.

[18] It should be clear that untargeted strategies are simply special cases of targeted strategies—constant functions, either $a_j(x) \equiv a_j$ or $p_j(x) \equiv p_j$ for $j = 1, 2$.

[19] To be more precise, this is except on a set of measure zero. We will gloss over this technicality in what follows.

[20] Indeed, in the Esteban et al. (2001) monopoly model, where price is not targeted, this is the basis for the optimality of advertising spending targeting.

[21] By contrast, in the Iyer et al. (2005) model, because loyalists and switchers are directly targetable, the price to loyalists can be different from the price to switchers.

[22] The targeting is implicit because as noted earlier, "loyalists" and "switchers" cannot be targeted explicitly.

[23] Now, the Thisse and Vives (1988) result can be seen as a special case of Proposition 1, namely part (2). Because consumers in their model are fully informed from the beginning, it is as if $a_1(x) = a_2(x) \equiv 1$. Hence, there is no informational differentiation, and the targeted price equilibrium they show is exactly the same as in part (2), namely $p_s = \Delta$, $p_w = 0$. In our model, however, as we will soon see, once the firms' advertising strategies are considered, part (2) of Proposition 1 will never arise.

[24] This turns out to be the critical distinction between the simultaneous and sequential game forms. In the latter, Firm j, when choosing a_j, takes a_{3-j} as given but not $\underline{p}_{3-j}$. Then, $\underline{p}_j$ is not fixed either. As we show in the online appendix, the result is that now one firm behaves as a price-*setting* monopolist, whereas the other firm behaves as a price-taking monopolist.

[25] $v_s(x) > 4(0.5 - x)k \iff (r - tx)/(2(1 - 2x)) > k$. The left-hand side is increasing in x; hence, its minimum value is $r/2$, and the observation follows.

[26] To see this, compare, for a given awareness level a, the weaker firm's lowest price in the full targeting equilibrium with the stronger firm's lowest price in the partial price targeting equilibrium. The former is $v_w(x) - v_s(x)a$, whereas the latter is $v_s(x) - v_w(x)a$.

[27] This joint optimality breaks down if there is even a small incremental cost of expanding the scope of targeting—no matter how small. Then, only partial advertising targeting would be optimal.

[28] If $a_1^{UA} = a_2^{UA} = 1$, then $\underline{p}_s^{UA}(x) \equiv \Delta(x)$, $\underline{p}_w^{UA}(x) \equiv 0$, and $\int_0^1 \underline{p}_1^{UA}(x) dx = \int_0^1 \underline{p}_2^{UA}(x) dx$, a contradiction.

References

Anderson SP, De Palma A, Thisse J-F (1992) *Discrete Choice Theory of Product Differentiation* (MIT Press, Cambridge, MA).

Brahim NBE-B, Lahmandi-Ayed R, Laussel D (2011) Is targeted advertising always beneficial? *Internat. J. Indust. Organ.* 29(6):678–689.

Bruell A (2017) TV's next act: Targeting ads at yogurt lovers and home buyers. *Wall Street Journal* (June 18), https://www.wsj.com/articles/tvs-next-act-targeting-ads-at-yogurt-lovers-and-home-buyers-1497778200.

Bruell A (2019) Targeted TV advertising is taking off. *Wall Street Journal* (June 16), https://www.wsj.com/articles/targeted-tv-advertising-is-taking-off-11560707549.

Bruell A (2021) Google's ad changes prompt big brands to revamp data strategies. *Wall Street Journal* (April 1), https://www.wsj.com/articles/googles-ad-changes-prompt-big-brands-to-revamp-data-strategies-11617269400#:~:text=To%20replace%20individual%20ad%20targeting,party%20data%20to%20target%20individuals.

Butters GR (1977) Equilibrium distributions of sales and advertising prices. *Rev. Econom. Stud.* 44(3):465–491.

Chandra A (2021) Mobile acquisition on location: 3 brands using geo-conquesting to win new customers. *ANA*. Accessed May 8, 2022, https://www.ana.net/blogs/show/id/mm-blog-2021-02-using-geo-conquesting-to-win-customers.

Chen Y, Iyer G (2002) Research note consumer addressability and customized pricing. *Marketing Sci.* 21(2):197–208.

Christou C, Vettas N (2008) On informative advertising and product differentiation. *Internat. J. Indust. Organ.* 26(1):92–112.

Clark CR, Doraszelski U, Draganska M (2009) The effect of advertising on brand awareness and perceived quality: An empirical investigation using panel data. *QME* 7(2):207–236.

Corts KS (1998) Third-degree price discrimination in oligopoly: All-out competition and strategic commitment. *RAND J. Econom.* 29(2):306–323.

d'Aspremont C, Gabszewicz JJ, Thisse J-F (1979) On Hotelling's stability in competition. *Econometrica* 47(5):1145–1150.

Dorfman R, Steiner PO (1954) Optimal advertising and optimal quality. *Amer. Econom. Rev.* 44(5):826–836.

Dubé J-P, Fang Z, Fong N, Luo X (2017) Competitive price targeting with smartphone coupons. *Marketing Sci.* 36(6):944–975.

Dwoskin E (2014) Pandora thinks it knows if you are a Republican. *Wall Street Journal* (February 13), https://www.wsj.com/articles/SB10001424052702304315004579381393567130078.

Esteban L, Gil A, Hernández JM (2001) Informative advertising and optimal targeting in a monopoly. *J. Indust. Econom.* 49(2):161–180.

Esteves R-B, Resende J (2016) Competitive targeted advertising with price discrimination. *Marketing Sci.* 35(4):576–587.

Fong NM, Fang Z, Luo X (2015) Geo-conquesting: Competitive locational targeting of mobile promotions. *J. Marketing Res.* 52(5): 726–735.

Goeree MS (2008) Limited information and advertising in the U.S. personal computer industry. *Econometrica* 76(5):1017–1074.

Gordon BR, Zettelmeyer F, Bhargava N, Chapsky D (2019) A comparison of approaches to advertising measurement: Evidence from big field experiments at Facebook. *Marketing Sci.* 38(2):193–225.

Grossman GM, Shapiro C (1984) Informative advertising with differentiated products. *Rev. Econom. Stud.* 51(1):63–81.

Haggin P, Vranica S (2021) Apple's moves to tighten flow of user data leave advertisers anxious. *Wall Street Journal* (June 9), https://www.wsj.com/articles/apples-moves-to-tighten-flow-of-user-data-leave-advertisers-anxious-11623189903.

Hotelling H (1929) Stability in competition. *Econom. J.* 39(153):41–57.

Iyer G, Soberman D, Miguel Villas-Boas J (2005) The targeting of advertising. *Marketing Sci.* 24(3):461–476.

Lobschat L, Osinga EC, Reinartz WJ (2017) What happens online stays online? Segment-specific online and offline effects of banner advertisements. *J. Marketing Res.* 54(6):901–913.

McAfee RP (1994) Endogenous availability, cartels, and merger in an equilibrium price dispersion. *J. Econom. Theory* 62(1):24–47.

Nair HS, Misra S, Hornbuckle WJ, Mishra R, Acharya A (2017) Big data and marketing analytics in gaming: Combining empirical models and field experimentation. *Marketing Sci.* 36(5):699–725.

Narasimhan C (1988) Competitive promotional strategies. *J. Bus.* 61(4):427–449.

O'Connor P (2014) Political ads take targeting to the next level. *Wall Street Journal* (July 14), https://www.wsj.com/articles/political-ads-take-targeting-to-the-next-level-1405381606.

Pazgal A, Soberman D, Thomadsen R (2022) Consumer informedness: A key driver of differentiation. *J. Econom. Management Strategy* 31(2):356–368.

Rhodes A, Zhou J (2022) Personalized Pricing and privacy choice. Working Paper No. 1333, Toulouse School of Economics, Toulouse, France.

Sahni NS, Zou D, Chintagunta PK (2016) Do targeted discount offers serve as advertising? Evidence from 70 field experiments. *Management Sci.* 63(8):2688–2705.

Shaffer G, Zhang ZJ (1995) Competitive coupon targeting. *Marketing Sci.* 14(4):395–416.

Shaffer G, Zhang ZJ (2002) Competitive one-to-one promotions. *Management Sci.* 48(9):1143–1160.

Sharp B (2010) *How Brands Grow* (Oxford University Press, South Melbourne, VIC, Australia).

Solomon Partners (2021) Media monthly: Report summarizing select media & tech activity. Accessed, https://solomonpartners.com/2021/03/19/media-monthly-march2021/.

Stegeman M (1991) Advertising in competitive markets. *Amer. Econom. Rev.* 81(1):210–223.

Thisse J-F, Vives X (1988) On the strategic choice of spatial price policy. *Amer. Econom. Rev.* 78(1):122–137.

Tirole J (1988) *The Theory of Industrial Organization* (MIT Press, Cambridge, MA).

Vascellaro JE (2011) TV's next wave: Tuning in to you. *Wall Street Journal* (March 7), https://www.wsj.com/articles/SB10001424052748704288304576171251689944350.

Vilcassim NJ, Kadiyali V, Chintagunta PK (1999) Investigating dynamic multifirm market interactions in price and advertising. *Management Sci.* 45(4):499–518.